Lecture 6 Practice List: Detailed Solutions

Economics of Organizations and Incentives (MPE Insper)

Solutions to the 28 exercises of `Listas/pratica-lecture-06.md`

November 2026

Scope. Lecture 6 is estimated (no slides yet): bargaining (announced for Lecture 5 but absent from its slides), tacit collusion, reputation and relational contracts. The items follow the order of the list: Tier 1 Foundations (1–8), Tier 2 Exam level (9–18), Tier 3 Advanced (19–28). Each solution has the same layout: what is asked, method, solution, interpretation, verification and common mistakes, followed by a boxed answer. Items 12 (Tadelis 11.5) and 19 (Gibbons 2.20) are the solutions of `Listas/lista-reais-lectures-01-05-solutions/`, reused unchanged. Every number is asserted by `check_T1.py`, `check_T2.py` and `check_T3.py` (one per tier). Where Muñoz-Garcia prints an answer, ours was cross-checked: the printed answer to Ch. 6 Ex. 3(c) (item 13) contradicts the book's own numbers and is flagged there.

Contents

Tier 1 — Foundations **2**

Item 1: Muñoz-Garcia Ch. 5, Exercise 4 (p. 228) **2**

Item 2: Cabral Exercise 9.1 (p. 242) **4**

Item 3: Tadelis Exercise 11.1 (p. 236) **6**

Item 4: Muñoz-Garcia Ch. 4, Exercise 1 (p. 154) **8**

Item 5: Milgrom & Roberts Ch. 8, Mathematical Exercise 4 (p. 286) **10**

Item 6: Milgrom & Roberts Ch. 8, Food for Thought 3 (p. 284) **12**

Item 7: Milgrom & Roberts Ch. 8, Food for Thought 4 (p. 284–285) **14**

Item 8: Tadelis Exercise 11.3 (p. 236) **16**

Tier 2 — Exam level **19**

Item 9: Tadelis Exercise 10.8 (p. 218) **19**

Item 10: Tadelis Exercise 11.2 (p. 236) **21**

Item 11: Gibbons Problem 2.19 (p. 137) **23**

Item 12: Tadelis Exercise 11.5 (p. 236–237) **25**

Item 13: Muñoz-Garcia Ch. 6, Exercise 3 (p. 282–283) **28**

Item 14: Gibbons Problem 2.15 (p. 136) **30**

Item 15: Tadelis Exercise 10.4 (p. 216) **32**

Item 16: Milgrom & Roberts Ch. 8, Mathematical Exercise 5 (p. 286) **34**

Item 17: Milgrom & Roberts Ch. 8, Mathematical Exercise 3 (p. 286) **36**

Item 18: Tadelis Exercise 10.6 (p. 217) **38**

Tier 3 — Advanced **40**

Item 19: Gibbons Problem 2.20 (p. 137–138) **40**

Item 20: Muñoz-Garcia Ch. 6, Exercise 5 (p. 290–291) **43**

Item 21: Cabral Exercise 9.15 (p. 244) **45**

Item 22: Muñoz-Garcia Ch. 6, Exercise 10 (p. 310) **47**

Item 23: Gibbons Problem 2.14 (p. 135–136) **49**

Item 24: Tadelis Exercise 11.7 (p. 237) **51**

Item 25: Cabral Exercise 9.19 (p. 245) **55**

Item 26: Tadelis Exercise 10.10 (p. 219) **57**

Item 27: Gibbons Problem 2.17 (p. 136–137) **59**

Item 28: Milgrom & Roberts Ch. 8, Mathematical Exercise 1 (p. 285) **61**

Tier 1 — Foundations

Item 1: Muñoz-Garcia Ch. 5, Exercise 4

Foundations · difficulty 2/5

Statement (verbatim from the list, item 1)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 5, Exercise 4 "Cartel with Firms Competing a la Cournot—Unsustainable in a One-Shot game", p. 228 [PDF 240]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider two firms competing in output levels (a la Cournot) and facing linear inverse demand $p(Q) = 100 - Q$, where $Q = q_1 + q_2$ denotes aggregate output. For simplicity, assume that firms face a common marginal cost of production $c = 10$.
>
> (a) *Best response function.* Set up each firm's profit-maximization problem and find its best response function. Interpret.
>
> (b) *Cournot competition.* Find the equilibrium output each firm produces when competing a la Cournot (that is, when they simultaneously and independently choose their output levels). In addition, find the profits that each firm earns in equilibrium.
>
> (c) *Collusive agreement.* Assume now that the CEOs from both companies meet for lunch and start talking about how they could increase their profits if they could increase market prices by reducing their production. That is, firms are trying to collude! Set up the maximization problem that firms solve when maximizing their *joint* profits (that is, the sum of profits for both firms). Find the output level that each firm should select to maximize joint profits. In addition, find the profits that each firm obtains in this collusive agreement.
>
> (d) *Profit comparison.* Compare the profits that firms obtain when competing a la Cournot (from part b) against their profits when they successfully collude (from part c).
>
> (e) *Unsustainable cartel.* Let us now show that the collusive agreement from part (c) cannot be sustained as a Nash equilibrium of the one-shot game, that is, when firms interact only once. Using the best response function found in part (a), show that when firm i chooses the collusive output found in part (c) its rival (firm j) responds with an output that is different from its collusive output. Interpret.

(a) What is asked. A Cournot duopoly with $p = 100 - Q$ and $c = 10$: (a) the best-response function; (b) the Cournot output and profit; (c) the joint-profit-maximising (collusive) outputs and profits; (d) the comparison; (e) why the collusive profile is not a Nash equilibrium of the one-shot game.

(b) Method. Solve each first-order condition. A cartel maximises the *sum* of profits. A Nash equilibrium requires every output to be a best reply, so we test the collusive output against the best-response function of (a).

(c) Solution. **(a) Best response.** Firm i solves $\max_{q_i} (100 - q_i - q_j)\, q_i - 10 q_i$. The first-order condition $100 - 2q_i - q_j - 10 = 0$ gives

$$q_i(q_j) = 45 - \tfrac{1}{2}\, q_j.$$

Firm i cuts output as j expands (strategic substitutes, slope $-\frac{1}{2}$).

(b) Cournot. Setting $q_1 = q_2 = q$ in (a): $q = 45 - q/2$, so $q^C = 30$. Price is $100 - 60 = 40$ and profit is $(40 - 10) \cdot 30 = 900$ for each firm (1 800 in total).

(c) Collusion. The cartel maximises $(100 - Q - 10)Q$. The condition $90 - 2Q = 0$ gives $Q^m = 45$, i.e. $q^m = 22.5$ each. Price is 55 and each firm earns $(55 - 10) \cdot 22.5 = 1\,012.5$ (2 025 in total).

(d) Comparison. $1\,012.5 > 900$: collusion raises each profit by 112.5 (12.5%), by restricting output from 60 to 45 and lifting the price from 40 to 55.

(e) No Nash equilibrium in the one-shot game. If firm i produces $q^m = 22.5$, firm j's best reply is $q_j(22.5) = 45 - 11.25 = 33.75 \neq 22.5$. Its profit from this reply is $(100 - 56.25 - 10) \cdot 33.75 = 1\,139.0625 > 1\,012.5$. Each firm has a profitable deviation, so $(22.5, 22.5)$ is not a Nash equilibrium. The only Nash equilibrium is $(30, 30)$.

(d) Interpretation. The cartel output is Pareto-superior for the firms but each firm's private incentive is to overproduce, a Prisoner's Dilemma. A one-shot promise to restrict output is not credible. Only repetition with a punishment can support it (item 9 for the cutoff; item 13 for the same numbers under temporary punishments).

(e) Verification. `check_T1.py` (item 1) solves the first-order conditions symbolically: $q^C = 30$, $\pi^C = 900$, $Q^m = 45$, $\pi^m = 1\,012.5$, the reply 33.75 and the deviation profit $1\,139.0625 = \frac{18225}{16}$, and checks on a grid of step $\frac{1}{100}$ that 30 is the best reply to 30. The book's answer (worked after the exercise) gives the same numbers.

(f) Common mistakes.

- Maximising each firm's own profit in (c). The cartel maximises the sum, so each firm internalises the price effect on the other.
- Concluding in (e) that firm j "joins" the agreement because both gain. The test is a unilateral best reply, not joint profit.

Answer. (a) $q_i = 45 - q_j/2$. (b) $q^C = 30$, $\pi^C = 900$. (c) $q^m = 22.5$, $\pi^m = 1\,012.5$. (d) Collusion pays more ($1\,012.5 > 900$). (e) The best reply to 22.5 is 33.75, worth $1\,139.06 > 1\,012.5$, so the cartel is not a one-shot Nash equilibrium.

Item 2: Cabral Exercise 9.1

Foundations · difficulty 1/5

Statement (verbatim from the list, item 2)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 9 Review and practice exercises, Exercise 9.1 "Tacit collusion", p. 242 [PDF 254]. Reproduced verbatim. Answer in the source: none in the book.

> **9.1. Tacit collusion.** Industries A and B can be characterized by a series of parameters: n, the number of firms, is 8 in both industries; r, the annual interest rate, is 10% in both countries; f, the frequency with which firms interact (number of times per year), is 1 in industry A and 12 in industry B; g, the industry growth rate, is 10% in industry A and -30% in industry B; finally, h, the likelihood that the industry will continue in existence into the next period, is 80% in industry A and 100% in industry B.
>
> In which of the two industries do you think tacit collusion is more likely to take place? Briefly justify your answer.

(a) What is asked. Two industries with $n = 8$ firms and $r = 10\%$ differ in the frequency f of interaction, the growth rate g and the survival probability $1 - h$. In which is tacit collusion more likely?

(b) Method. Collusion by grim trigger holds iff the effective discount factor is high enough (Cabral §9.1, pp. 218–222). Cabral's expression (9.5) is $\delta = \dfrac{(1+g)(1-h)}{1+r/f}$: δ rises with f and g and falls with h and r. With n symmetric firms the monopoly price is sustained iff $\delta \geq 1 - \frac{1}{n}$ (Cabral p. 227, the n-firm version of (9.4)).

(c) Solution. **Industry A** ($f = 1$, $g = 10\%$, continuation 80% so $h = 20\%$):

$$\delta_A = \frac{1.1 \times 0.8}{1 + 0.10/1} = \frac{0.88}{1.10} = 0.80.$$

Industry B ($f = 12$, $g = -30\%$, continuation 100% so $h = 0$):

$$\delta_B = \frac{0.7 \times 1}{1 + 0.10/12} = \frac{0.7}{1.00833} \approx 0.694.$$

Threshold. For $n = 8$: $1 - \frac{1}{8} = 0.875$. Both δ_A and δ_B fall short, so full monopoly pricing is sustainable in neither. Comparing them, $\delta_A = 0.80 > \delta_B = 0.69$, so industry A is closer to the threshold and collusion is more likely there. The fast frequency of B ($f = 12$) helps, but the market shrinks by 30% a year, so tomorrow's profits are worth little, and this dominates the benefit of frequent interaction.

Reading of g. Formula (9.5) takes g and h as per-period values. If instead -30% is an *annual* rate with monthly interaction, the monthly growth factor is $0.7^{1/12} \approx 0.971$ and $\delta_B \approx 0.971/1.00833 \approx 0.963 > 0.875$: the ranking reverses and B would sustain collusion. The exercise lists g next to r (annual) and applies f only to r, so we follow (9.5) as written and flag the dependence.

(d) Interpretation. The discount factor combines three forces: impatience (r/f), growth and survival. An industry that is declining or likely to disappear (pharmaceuticals with fast obsolescence; Cabral's example) cannot hold a cartel together even when firms meet often, since the cost of a price war is paid in the future. Eight firms make the cutoff demanding (0.875): the profit from undercutting is the whole market, whereas the cartel profit is $\frac{1}{8}$ of it.

(e) Verification. `check_T1.py` computes $\delta_A = \frac{4}{5}$, $\delta_B = 0.6942$, the cutoff $\frac{7}{8}$, the alternative reading (0.9627) and the monotonicity of (9.5) in f, g, h. Cabral prints no answer.

Answer. $\delta_A = 0.80$ and $\delta_B = 0.694$; the cutoff for $n = 8$ is 0.875. Neither industry sustains full collusion at (9.5) as written, but A (higher δ) is the more likely. Reading the -30% as annual would reverse this ($\delta_B = 0.963$).

Item 3: Tadelis Exercise 11.1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 3)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 11.1 "Disagreement", p. 236 [PDF 255]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Disagreement:** Construct a pair of strategies for the ultimatum game ($T = 1$ bargaining game) that constitute a Nash equilibrium and together support the outcome that there is no agreement reached by the two players and the payoffs are zero to each. Show that this disagreement outcome can be supported by a Nash equilibrium regardless of the number of bargaining periods.

(a) What is asked. Construct a Nash equilibrium of the ultimatum game ($T = 1$) in which the players disagree and both get zero, and show that disagreement is a Nash outcome for every number of bargaining periods.

(b) Method. Nash equilibrium needs only best replies on the equilibrium path, so a threat to reject every offer is allowed (Tadelis §11.1, pp. 220–224). Subgame perfection removes such threats because rejecting a positive offer is not a best reply in the last period (§11.2, pp. 224–227).

(c) Solution. Ultimatum ($T = 1$). Player 1 offers $x_1 = 1$ for himself (player 2 gets 0), and player 2 rejects every offer. Outcome: rejection, payoffs $(0, 0)$.

- *Player 1:* any other offer is rejected too, so he still gets 0. No deviation gains.
- *Player 2:* given the offer $(1, 0)$ she gets 0 whether she rejects or accepts, so rejecting is a best reply (weakly).

So this is a Nash equilibrium with payoffs $(0, 0)$.

T periods. Let every proposer claim the whole pie and every responder reject every offer in every period. The path runs through all T periods without agreement: payoffs $(0, 0)$. A unilateral deviation cannot yield positive payoff: as proposer the deviator's offer (whatever it is) is rejected; as responder he can only accept an offer that leaves him $\delta^{t-1} \cdot 0 = 0$, because the opponent claims the whole pie. Hence no one gains, for every T and every δ.

Why not subgame perfect. In the last period, a responder offered $x > 0$ gets x by accepting and 0 by rejecting. The strategy "reject everything" is therefore not a best reply in that subgame (it is a non-credible threat), so the profile fails subgame perfection. Backward induction rules it out and selects immediate agreement (item 8 for finite T; Tadelis §11.3 for the infinite horizon).

(d) Interpretation. Nash equilibrium alone has no predictive power in bargaining: any division, and also disagreement, can be supported by threats to reject anything else. This is why the course uses SPE.

(e) Verification. `check_T1.py` (item 3) evaluates the deviator's best payoff against "claim everything / reject everything" for $T = 1, \ldots, 6$ and $\delta \in \{\frac{1}{2}, \frac{9}{10}\}$ and finds 0, equal to the equilibrium payoff. Tadelis prints no answer.

(f) Common mistakes.

- Calling the profile subgame perfect because no one deviates on the path. The test also applies after offers that are never made.
- Letting player 1 offer player 2 a positive share. She would then gain by accepting, so rejecting would not even be a best reply on the path. The offer $(1, 0)$ makes her indifferent.

Answer. Player 1 offers himself the whole pie, player 2 rejects every offer. This is Nash with payoffs $(0, 0)$: no deviation yields a positive payoff. The same strategies (claim everything; reject everything) in every period work for any T. The profile is not subgame perfect because rejecting a positive offer in the last period is not a best reply.

Item 4: Muñoz-Garcia Ch. 4, Exercise 1

Foundations · difficulty 2/5

Statement (verbatim from the list, item 4)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 4, Exercise 1 "Ultimatum Bargaining Game", p. 154 [PDF 166]. Reproduced verbatim (statement only; the book's worked answer follows it).

> In the ultimatum bargaining game, a proposer is given a pie, normalized to a size \$1, and he is asked to make a monetary offer, x, to the responder who, upon receiving the offer, only has the option to accept or reject it (as if he received an "ultimatum" from the proposer). If the offer x is accepted, then the responder receives it while the proposer keeps the remainder of the pie $1 - x$. However, if he rejects it, both players receive a zero payoff. Operating by backward induction, the responder should accept any offer x from the proposer (even if it is low) since the alternative (reject the offer) yields an even lower payoff (zero). Anticipating such as a response, the proposer should then offer one cent (or the smallest monetary amount) to the responder, since by doing so the proposer guarantees acceptance and maximizes his own payoff. Therefore, according to the subgame perfect equilibrium prediction in the ultimatum bargaining game, the proposer should make a tiny offer (one cent or, if possible, an amount approaching zero), and the responder should accept it, since his alternative (reject the offer) would give him a zero payoff. However, in experimental tests of the ultimatum bargaining game, subjects who are assigned the role of proposer rarely make offers close to zero to the subject who plays as a responder.
>
> Furthermore, sometimes subjects in the role of the responder reject positive offers, which seems to contradict our equilibrium predictions. In order to explain this dissonance between theory and experiments, many scholars have suggested that players' utility function is not as selfish as that specified in standard models (where players only care about the monetary payoff they receive). Instead, the utility function should also include social preferences, measured by the difference between the payoff a player obtains and that of his opponent, which gives rise to envy (when the monetary amount he receives is lower than that of his opponent) or guilt (when his monetary payoff is higher than his opponent's). In particular, suppose that the responder's utility is given by
>
> $$u_R(x, y) = x + \alpha(x - y),$$
>
> where x is the responder's monetary payoff, y is the proposer's monetary payoff, and α is a positive constant. That is, the responder not only cares about how much money he receives, x, but also about the payoff inequality that emerges at the end of the game, $\alpha(x - y)$, which gives rise to either envy, if $x < y$, or guilt, if $x > y$. For simplicity, assume that the proposer is selfish, i.e., his utility function only considers his own monetary payoffs $u_P(x, y) = y$, as in the basic model.
>
> **Part (a)** Use a game tree to represent this game in its extensive form, writing the payoffs in terms of m, the monetary offer of the proposer, and parameter α.
>
> **Part (b)** Find the subgame perfect equilibrium. Describe how equilibrium payoffs are affected by changes in parameter α.
>
> **Part (c)** Depict the equilibrium monetary amount that the proposer keeps, and the payoff that the responder receives, as a function of parameter α.

(a) What is asked. An ultimatum game in which the responder's utility is $u_R = x + \alpha(x - y)$ (x his money, y the proposer's, $\alpha > 0$) and the proposer is selfish: (a) the game tree; (b) the SPE and how it depends on α; (c) the proposer's and the responder's equilibrium amounts as functions of α.

(b) Method. Backward induction. Let m be the offer to the responder, so $x = m$ and $y = 1 - m$. The responder accepts iff accepting is at least as good as rejecting, which gives 0 (with $x = y = 0$).

(c) Solution. **(a) Tree.** Player 1 (proposer) chooses $m \in [0, 1]$. Player 2 (responder) then accepts, with payoffs $\big(1 - m,\ m + \alpha(2m - 1)\big)$, or rejects, with payoffs $(0, 0)$. The responder's payoff from acceptance is $m + \alpha\big(m - (1 - m)\big) = m + \alpha(2m - 1)$.

(b) SPE. The responder accepts iff

$$m + \alpha(2m - 1) \geq 0 \iff m \geq \frac{\alpha}{1 + 2\alpha} \equiv m^*.$$

The proposer's payoff $1 - m$ is decreasing in m, so he offers exactly m^*, which is accepted. Equilibrium payoffs:

$$(1 - m^*,\ m^*) = \Big(\frac{1 + \alpha}{1 + 2\alpha},\ \frac{\alpha}{1 + 2\alpha}\Big).$$

Strategies: proposer offers m^*; responder accepts iff $m \geq m^*$. Comparative statics:

$$\frac{\partial m^*}{\partial \alpha} = \frac{1}{(1 + 2\alpha)^2} > 0, \qquad \frac{\partial (1 - m^*)}{\partial \alpha} = -\frac{1}{(1 + 2\alpha)^2} < 0.$$

(c) Plot. At $\alpha = 0$ the split is $(1, 0)$, the standard SPE. At $\alpha = 0.5$ it is $(0.75, 0.25)$, at $\alpha = 1$ it is $(\frac{2}{3}, \frac{1}{3})$, and as $\alpha \to \infty$ both amounts tend to $\frac{1}{2}$. The proposer's curve falls from 1 towards $\frac{1}{2}$ and the responder's rises from 0 towards $\frac{1}{2}$ without crossing: $1 - 2m^* = \frac{1}{1+2\alpha} > 0$, so the proposer always keeps more.

(d) Interpretation. An aversion to inequality (here, envy when $x < y$) makes the responder demand a larger share, as in experiments where low offers are rejected. The theory stays an SPE of a game with different *utilities*: a rejection of a positive offer is rational once the responder cares about fairness.

(e) Verification. `check_T1.py` (item 4) solves $u_R = 0$ symbolically ($m^* = \alpha/(1 + 2\alpha)$), checks both derivatives, the values at $\alpha = \frac{1}{2}$ and $\alpha = 1$, the limit $\frac{1}{2}$ and the identity $1 - 2m^* = \frac{1}{1+2\alpha}$. Matches the book's answer (it reports $(0.75, 0.25)$ and $(2/3, 1/3)$).

Answer. The responder accepts iff $m \geq \alpha/(1+2\alpha)$. SPE: the proposer offers $m^* = \alpha/(1+2\alpha)$, accepted. Payoffs $(\frac{1+\alpha}{1+2\alpha}, \frac{\alpha}{1+2\alpha})$: the proposer's falls and the responder's rises with α, from $(1, 0)$ at $\alpha = 0$ towards $(\frac{1}{2}, \frac{1}{2})$ as $\alpha \to \infty$.

Item 5: Milgrom & Roberts Ch. 8, Mathematical Exercise 4Foundations · difficulty 2/5

Statement (verbatim from the list, item 5)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Mathematical Exercise 4, p. 286 [PDF 301]. Reproduced verbatim. Answer in the source: none in the book.

> 4. In the discussion of the game of offering trust in the text, it was shown that if the game is being played only once, then trust will not be offered and, if it should be offered, it will not be honored. Suppose instead the two parties know they will interact exactly twice. Would trust be offered and honored in the second, final interaction? Why or why not? If the parties anticipate this and trust is offered in the first round, will it be honored? Why or why not? [Hint: Will honoring trust affect behavior at the last round?] Will trust then be offered? By this logic, what would happen if the parties knew they would interact exactly three times? Any arbitrary, but finite and known number of times? Do you find this to be a credible prediction about how people actually would behave? Why or why not?

(a) What is asked. In the trust game of Ch. 8 (offeror, decision maker), the parties meet exactly twice, then exactly three times, then any known finite number of times. Would trust be offered and honoured? Is the prediction credible?

(b) Method. Backward induction from the last round. The stage payoffs of Ch. 8 (pp. 261–263) are: offer and honour (V, V); offer and not honour $(-L, V+G)$; no offer $(0,0)$, with $V, G, L > 0$. In one play, a self-interested decision maker does not honour $(V+G > V)$, so the offeror does not offer $(-L < 0)$.

(c) Solution. Two interactions. In round 2 (final), nothing follows, so the one-shot logic applies: trust is not honoured if offered, and so it is not offered. Round-2 payoffs are $(0,0)$ whatever happened in round 1. In round 1, honouring cannot earn a reputation, since round 2's outcome does not depend on round 1 (the hint's question). The decision maker therefore gains G by not honouring, so he will not honour, and the offeror, anticipating this, does not offer. Both parties get 0 in each round.

Three interactions and any finite N. By induction: suppose rounds $k+1, \ldots, N$ all give $(0,0)$ regardless of history. Then round k is effectively one-shot, with the same conclusion. The unravelling reaches round 1, so for every known finite N trust is never offered and, if it were, never honoured. The formal SPE is unique.

Credibility. The prediction is not credible as a description of behaviour. Experiments with finitely repeated trust and social-dilemma games show cooperation for most of the horizon, with collapse only near the end. Explanations: (i) people are unsure of the horizon (a probability of continuation restores trust, item 16); (ii) some players are honest by type, so a selfish player gains by mimicking them (reputation, Lectures 7–8, e.g. Kreps–Milgrom–Roberts–Wilson); (iii) a small departure from rationality (ε-equilibrium) sustains cooperation for long horizons.

(d) Interpretation. The finite-horizon paradox shows that trust rests on the *shadow of the future*: repetition disciplines behaviour only if the interaction has no known last round.

(e) Verification. `check_T1.py` (item 5) runs backward induction on the repeated trust game for $N = 1, 2, 3, 10$ (with $V = 3$, $G = 2$, $L = 5$): in every round the decision maker dishonours and the offeror does not offer, giving continuation $(0, 0)$. The exercise is discursive and has no printed answer.

Answer. Round N is one-shot, so trust is neither offered nor honoured; honouring in round $N - 1$ cannot buy a future reward; by induction the same holds in every round. For any known finite horizon no trust is offered. This is not credible empirically: with an uncertain end date, honest types or small departures from rationality, cooperation reappears.

Item 6: Milgrom & Roberts Ch. 8, Food for Thought 3 Foundations · difficulty 2/5

Statement (verbatim from the list, item 6)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Food for Thought 3, p. 284 [PDF 299]. Reproduced verbatim. Answer in the source: none in the book.

> 3. In many countries, corporations whose shares are publicly traded are required to have their financial statements audited by independent accountants who check whether the financial information being provided by management to investors is accurate and has been prepared following accepted methods and procedures and who then publicly attest to their findings. The auditors are generally chosen by management (perhaps subject to nominal stockholder approval). Audit work provides a major source of income for accounting firms. Generally, each accounting firm has many clients that it audits: Even when it might be technically feasible for a relatively small accounting firm to audit a large corporation's records, there seems to be a reluctance for corporations to use audit firms when a single client would represent too much of the accountant's business. Instead, a handful of extremely large accounting firms typically do almost all the auditing of large corporations, with each having many corporate clients. How do you account for this?

(a) What is asked. Why do large corporations use big audit firms with many clients, and avoid a firm for which one client would be too large a share of its business?

(b) Method. The chapter's reputation model (pp. 259–268; efficiency-wage condition, pp. 250–255). A party who can cheat for a one-time gain G stays honest iff the value of the reputation he would lose exceeds G: in the efficiency-wage form, $p\,(w-\bar{w})\,N \geq G$. We read the auditor's reputation as the trust-based rent v that investors' willingness to rely on its reports earns it. The answer is an analytical argument, not a computation; the chapter prints no answer to this question.

(c) Solution. 1. What auditors sell. Investors cannot verify the books themselves. An audit is valuable only if the auditor reports honestly even though management picks and pays him. This is the trust game of Ch. 8: investors offer trust, and the auditor can fail to honour it by shading the report to please management (gain G). Contracts cannot specify honest reporting in detail, so reputation has to enforce it ("Reputations as contract enforcers", p. 259).

2. Many clients make honesty cheaper to enforce.

- *Reputation is firm-wide and indivisible.* A shaded report for one client, once detected, damages the auditor's credibility with all clients and with investors, so the multiplier N applies to the whole stream of rents (compare the Maghribi traders, who punished cheaters group-wide, p. 252). With k clients paying a rent v each, honesty requires $p\,k\,v\,N \geq G_i$ for every client i. The left side grows with k, while the temptation G_i from any one client is only part of the firm's income.
- *A dominant client changes both sides.* If one client is a large share of revenue, its favour is worth more (G_i large), and the auditor has more to lose from displeasing that client than from being caught. The client can credibly threaten to leave and take the auditor's income with it: a firm that depends on one buyer faces the buyer's hold-up. With many clients, no single threat is decisive.

- *Reputation outlives individuals.* The brand passes from one partner to the next (Tadelis §10.5.3, pp. 207–208), and large firms monitor and discipline their own partners, which keeps the detection probability p high.

3. Why management goes along. Managers want the credibility of a clean opinion in order to raise capital, so they choose a firm whose reputation investors recognise. A small firm dependent on one client has no reputation to lose, so investors would discount its opinion and the audit would be worthless to the very manager who buys it.

4. Auxiliary reasons. Economies of scale, specialised knowledge, pooling of litigation risk and the capacity to audit a large corporation's worldwide records also favour big firms. They explain size, but not the reluctance to hire a firm with a large client share, which is the puzzle.

(d) Interpretation. Audit independence is bought by making the auditor's stake in its whole reputation large relative to the gain from any single favour. This is also why regulators limit the share of a firm's fees that may come from one audit client.

(e) Verification. `check_T1.py` (item 6) illustrates the inequality $p\,k\,v\,N \geq G$ with assumed numbers ($G = 10$, $p = \frac{1}{2}$, $v = 1$, $N = 6$): honesty needs $k \geq 4$ clients, and $k = 3$ fails. The numbers are illustrative, not from the book.

Answer. The auditor's reputation for honest reporting is a firm-wide asset. A detected shaded report loses all clients' rents, while the gain from favouring one client is limited. A firm dominated by one client has little to lose from displeasing it and much to lose by losing it, so its independence is weak and investors discount its opinion. Large audit firms with many clients make honesty incentive compatible, and management gains from the credibility.

Item 7: Milgrom & Roberts Ch. 8, Food for Thought 4 Foundations · difficulty 2/5

Statement (verbatim from the list, item 7)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Food for Thought 4, pp. 284–285 [PDF 299–300]. Reproduced verbatim. Answer in the source: none in the book.

> 4. The proper level of salaries for public officials is often a matter of debate. In many countries, public officials supplement their incomes by performing other duties. In the United States, Congressmen deliver lectures for pay and former bureaucrats cash in on their connections by becoming high-paid lobbyists. Recent U.S. proposals call for tying pay increases to limits on these activities. Discuss the efficacy of such a plan.

(a) What is asked. Discuss the efficacy of proposals that tie pay increases for public officials to limits on their outside earnings (lectures for pay; lobbying after leaving office).

(b) Method. The efficiency-wage model of Ch. 8 (pp. 250–255). An official does not sell favours iff $p\,(w-\bar{w})\,N \geq g$, where w is pay in office, $\bar{w}$ the best alternative if dismissed, p the probability of detecting misconduct, N the horizon multiplier and g the one-time gain from misconduct. The minimum deterrent pay is $w = \bar{w} + g/(Np)$ (formula 8.3). The exercise is discursive, so this is an analytical answer.

(c) Solution. 1. Where outside activities enter. Outside income plays two roles. (i) Lecture fees and later lobbying jobs from interest groups can be the *payment for favours*: they raise g, and they do not vanish if the official is found out, because the lobby values the same connections, so the penalty $w-\bar{w}$ falls ($\bar{w}$ rises). (ii) They are also a way of paying the official, which keeps the visible salary w low and the rent $w-\bar{w}$ of the job small. Both reduce $p(w-\bar{w})N-g$.

2. What the plan does. Tying a pay rise to a cap on outside activities (a) raises w and (b) cuts g and the outside component of $\bar{w}$. Both move the incentive constraint the right way. In the model the plan works iff

- the raise exceeds the outside income given up, otherwise $w-\bar{w}$ does not grow. With $p=0.2$, $N=4$, $\bar{w}=100$ and $g=80$ the deterrent salary is $w=200$. An official who earns 30 on the side needs only 170, and a raise of 20 in exchange for the ban does not deter;
- the limits are enforceable: p must apply to the new rules (disclosure; bans on post-office lobbying with credible penalties), otherwise the ban is cosmetic;
- the post-office restriction is credible: a delayed lobbying job is a reward that cheating officials *keep*, so only a binding cooling-off period reduces g.

3. Costs and caveats. (a) Participation: pay must still attract capable officials, whose alternatives are lucrative. A cap lowers total pay unless the salary rises enough, which is politically costly. (b) Not all outside activity is corrupt: lectures spread expertise, so a blunt ban destroys value. (c) Payments can be displaced to relatives or consulting firms, so monitoring (p) is the weak link, as in Macaulay's Lord Clive example (remote, powerful officials, low p). (d) A higher salary raises the rent of honest officials too.

4. Assessment. The plan is effective as a package: a salary increase that at least replaces the lost income, caps on activities that are payments for favours, and credible monitoring. A pay rise

without limits leaves g untouched. Limits without a pay rise lower $w - \bar{w}$ and invite cheating or exit.

(d) Interpretation. The proposal applies Clive's logic of liberal remuneration (p. 252) together with the monitoring side of the model: pay and monitoring are substitutes in $p(w - \bar{w})N \geq g$.

(e) Verification. `check_T1.py` (item 7) computes the deterrent salary 200, the salary 170 with 30 of outside income, shows that a +20 raise against a 30 ban fails to deter, and that a larger gain ($g = 120$) raises the required salary to 250. The numbers are illustrative.

Answer. Deterrence needs $p(w - \bar{w})N \geq g$. A package of a higher salary, limits on paid outside work and enforceable post-office rules raises the rent and cuts the gain from favours, so it can work. It fails if the raise is smaller than the income given up, if the limits cannot be monitored, or if the cooling-off rule is not credible.

Item 8: Tadelis Exercise 11.3

Foundations · difficulty 2/5

Statement (verbatim from the list, item 8)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 11.3 "Even/Odd Symmetry", p. 236 [PDF 255]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Even/Odd Symmetry:** In Section 11.2 we analyzed the alternating-offer bargaining game for a finite number of periods when T was odd. Repeat the analysis for T even.

Referred to in the statement. The alternating-offer bargaining game (Tadelis Ch. 11 opening, pp. 220–221 [PDF p. 239–240]) and its analysis for odd T in Section 11.2 (pp. 225–226 [PDF p. 244–245]) (reproduced from the book):

> The simple framework considers two players that need to split a "pie" (representing the surplus from an agreement or the gains from trade). The pie is assumed to have a total value that is normalized to equal 1, and then the parties bargain on how to split the pie. [...] we can summarize the game as follows.
>
> In the first round:
>
> - Player 1 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
> - Player 2 then chooses to accept, causing the game to end with payoffs $v_1 = x$ and $v_2 = 1-x$, or reject, causing the game to move to the second round.
>
> In the second round:
>
> - A share $1-\delta$ of the pie is dissipated (the pie is discounted with discount factor $0 < \delta < 1$).
> - Player 2 offers shares $(x, 1-x)$, where player 1 receives x and player 2 receives $1-x$.
> - Player 1 chooses to accept, causing the game to end with payoffs $v_1 = \delta x$ and $v_2 = \delta(1-x)$, or reject, causing the game to move to the third round.
>
> In the third round and beyond:
>
> - The game continues in the same way, where following rejection in an odd period player 2 gets to offer in the next even period, and vice versa.
> - Each period has further discounting of the pie, so that in period t the total pie is worth δ^{t-1}.
>
> From the specification of the bargaining procedure we see that once a player agrees to the offer made by his opponent, the game will end with an agreement. If instead the players keep rejecting offers then the game may continue either indefinitely or until some prespecified deadline is reached. Assume for now that the game ends in some prespecified final period T, which will be reached if the players did not reach an agreement in the $T-1$ periods that preceded T. Furthermore assume that period T is a "hard deadline," so that both players receive a payoff of zero if they fail to reach an agreement in this last period. [...]
>
> Consider the case with an odd number of rounds $T < \infty$, implying that player 1 has both the first-mover and last-mover advantages. The following backward induction argument applies:
>
> - In period T player 2 accepts any offer, so player 1 offers $x = 1$ and payoffs are $v_1 = \delta^{T-1}$ and $v_2 = 0$.
> - In period $T-1$ (even period, player 2 offers), by backward induction player 1 should accept anything resulting in a payoff of $v_1 \geq \delta^{T-1}$. If player 1 is offered x in period $T-1$ then $v_1 = \delta^{T-2}x$. This implies that in period $T-1$ player 1 will accept any $x \geq \delta$ and by backward induction player 2 should offer $x = \delta$, which yields player 1 a payoff of $v_1 = \delta \times \delta^{T-2} = \delta^{T-1}$. Payoffs are then $v_1 = \delta^{T-1}$ and $v_2 = \delta^{T-2}(1-\delta)$.

- In period $T-2$ (odd period), conditional on the analysis for $T-1$, player 2's best response is to accept any x that gives him $\delta^{T-3}(1-x) \geq \delta^{T-2}(1-\delta)$. Player 1's best response to this is to offer the largest x that satisfies this inequality, and solving it with equality yields player 1's best response: $x = 1-\delta+\delta^2$. This offer followed by player 2's acceptance yields $v_1 = \delta^{T-3}x = \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$ and $v_2 = \delta^{T-2} - \delta^{T-1}$.
- In period $T-3$ (even period), using the same backward induction argument, player 1's best response is to accept any x that satisfies $\delta^{T-4}x \geq \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$, and player 2 then makes an offer that satisfies this constraint with equality, $x = \delta - \delta^2 + \delta^3$, yielding $v_1 = \delta^{T-3} - \delta^{T-2} + \delta^{T-1}$ and $v_2 = \delta^{T-4}(1-x) = \delta^{T-4} - \delta^{T-3} + \delta^{T-2} - \delta^{T-1}$.

We can continue with this tedious exercise only to realize that a simple pattern emerges. If we consider the solution for an odd period $T-s$ (s being even because T is assumed to be odd) then the backward induction argument leads to the sequentially rational offer

$$x_{T-s} = 1 - \delta + \delta^2 - \delta^3 + \cdots + \delta^s,$$

while for an even period $T-s$ (s being odd) then the backward induction argument leads to the sequentially rational offer

$$x_{T-s} = \delta - \delta^2 + \delta^3 - \delta^4 + \cdots + \delta^s.$$

We can use this pattern to solve for the subgame-perfect equilibrium offer in the first period, x_1, which by backward induction must be accepted by player 2 and it is equal to

$$\begin{aligned} x_1 &= 1 - \delta + \delta^2 - \delta^3 + \delta^4 + \cdots + \delta^{T-1} \\ &= (1 + \delta^2 + \delta^4 + \cdots + \delta^{T-1}) - (\delta + \delta^3 + \delta^5 + \cdots + \delta^{T-2}) \\ &= \frac{1-\delta^{T+1}}{1-\delta^2} - \frac{\delta - \delta^T}{1-\delta^2} \\ &= \frac{1+\delta^T}{1+\delta}, \end{aligned}$$

and this in turn implies that

$$v_1^* = x_1 = \frac{1+\delta^T}{1+\delta} \quad \text{and} \quad v_2^* = (1-x_1) = \frac{\delta - \delta^T}{1+\delta}. \tag{11.1}$$

(a) What is asked. Solve the finite alternating-offer bargaining game of Section 11.2 when the number of periods T is even.

(b) Method. Backward induction with a common discount factor δ (Tadelis §11.2, pp. 224–227). Player 1 proposes in odd periods and player 2 in even periods, so for T even *player 2 makes the last offer*. Let s_m be the share of the pie, valued at the start of the subgame, that the proposer obtains when m periods remain.

(c) Solution. Last period. The proposer takes the whole pie: $s_1 = 1$ (the responder accepts at indifference).

Earlier periods. With m periods left, the responder gets δs_{m-1} by rejecting, since next period he is the proposer with $m-1$ periods left. The proposer offers him exactly that and keeps

$$s_m = 1 - \delta\, s_{m-1}, \qquad s_1 = 1 \quad \Longrightarrow \quad s_m = \frac{1-(-\delta)^m}{1+\delta}.$$

(Induction: $s_m = 1 - \delta\,\frac{1-(-\delta)^{m-1}}{1+\delta} = \frac{1+\delta(-\delta)^{m-1}}{1+\delta}$ and $\delta(-\delta)^{m-1} = -(-\delta)^m$.)

T even. Player 1 is the proposer when T periods remain. His offer is accepted in period 1 (agreement is immediate, as in Prop. 11.3, p. 226):

$$(v_1, v_2) = \Big(\frac{1-\delta^T}{1+\delta},\ \frac{\delta+\delta^T}{1+\delta}\Big).$$

For $T = 2$: $v_1 = 1-\delta$ and $v_2 = \delta$. Strategies: with m periods left the proposer offers the responder δs_{m-1}, and the responder accepts iff his share is at least that.

Comparison with T odd (the book's case, $v_1 = \frac{1+\delta^T}{1+\delta}$): for T even the first mover's share is *below* $\frac{1}{1+\delta}$, and for T odd it is above. Both tend to the infinite-horizon share $\frac{1}{1+\delta}$ (§11.3) as $T \to \infty$. Player 2, who has the last word when T is even, is ahead ($v_2 > \frac{1}{2}$) iff $2\delta^T > 1-\delta$. Example: $T = 2$ and $\delta = 0.8$ gives $v_1 = 0.2$, $v_2 = 0.8$; with $\delta = 0.4$, $v_1 = 0.6$.

(d) Interpretation. The last word is worth a lot when patience is high ($\delta \to 1$): the last proposer can claim almost the whole pie, so the earlier proposer must buy agreement with the value of that position. Being first is an advantage only when delay is costly.

(e) Verification. `check_T1.py` (item 8) checks the recursion against the closed form symbolically for $T = 1, \ldots, 8$, the case $T = 2$, the equivalence "player 2 ahead $\iff 2\delta^T > 1-\delta$" on a grid, and the limit $\frac{1}{1+\delta}$ with the ordering $\frac{1-\delta^4}{1+\delta} < \frac{1}{1+\delta} < \frac{1+\delta^5}{1+\delta}$ (at $\delta = \frac{1}{2}$). Tadelis prints no answer.

(f) Common mistakes.

- Using the odd-T formula $\frac{1+\delta^T}{1+\delta}$. Parity decides who proposes last.
- Discounting with the wrong factor or from the wrong period: each earlier step uses δs_{m-1}, the responder's own continuation.

Answer. $s_m = \frac{1-(-\delta)^m}{1+\delta}$. For T even, agreement in period 1 with $(v_1, v_2) = (\frac{1-\delta^T}{1+\delta}, \frac{\delta+\delta^T}{1+\delta})$; e.g. $T = 2$ gives $(1-\delta, \delta)$. The last mover (player 2) is ahead iff $2\delta^T > 1-\delta$. The limit as $T \to \infty$ is $\frac{1}{1+\delta}$.

Tier 2 — Exam level

Item 9: Tadelis Exercise 10.8

Exam level · difficulty 3/5

Statement (verbatim from the list, item 9)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.8 "Tacit Collusion", p. 218 [PDF 237]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Tacit Collusion:** Two firms, which have zero marginal cost and no fixed cost, produce some good, each producing $q_i \geq 0$, $i \in \{1, 2\}$. The demand for this good is given by $p = 200 - Q$, where $Q = q_1 + q_2$.
>
> a. First consider the case of Cournot competition, in which each form chooses q_i and this game is infinitely repeated with a discount factor $\delta < 1$. Solve for the static stage-game Cournot-Nash equilibrium.
>
> b. For which values of δ can you support the firms' equally splitting monopoly profits in each period as a subgame-perfect equilibrium that uses grim-trigger strategies (i.e., after one deviates from the proposed split, they resort to the static Cournot-Nash equilibrium thereafter)? (Note: Be careful in defining the strategies of the firms!)
>
> c. Now assume that the firms compete à la Bertrand, each choosing a price $p_i \geq 0$, where the lowest-price firm gets all the demand and in case of a tie they split the market. Solve for the static stage-game Bertrand-Nash equilibrium.
>
> d. For which values of δ can you support the firms' equally splitting monopoly profits in each period as a subgame-perfect equilibrium that uses grim-trigger strategies (i.e., after one deviates from the proposed split, they resort to the static Bertrand-Nash equilibrium thereafter)? (Note: Be careful in defining the strategies of the firms!)
>
> e. Now instead of using grim-trigger strategies, try to support the firms' equally splitting monopoly profits as a subgame-perfect equilibrium in which after a deviation firms would resort to the static Bertrand competition for only two periods. For which values of δ will this work? Why is this answer different than your answer in (d)?

(a) What is asked. Two firms with zero cost face $p = 200 - Q$. (a) The static Cournot equilibrium. (b) The δ for which splitting monopoly profit is a grim-trigger SPE under Cournot. (c) The static Bertrand equilibrium. (d) The same cutoff under Bertrand. (e) A two-period Bertrand punishment: for which δ, and why does it differ from (d)?

(b) Method. One-stage deviation with grim trigger (Tadelis §10.4, pp. 201–204): cooperation is optimal iff $\dfrac{\pi^m}{1-\delta} \geq \pi^D + \dfrac{\delta\,\pi^N}{1-\delta}$, with π^m each firm's share of monopoly profit, π^D the best one-period deviation profit and π^N the static Nash profit. The punishment phase is a repeated static Nash equilibrium, so it needs no further check.

(c) Solution. **(a) Cournot.** Firm i maximises $(200 - q_i - q_j)q_i$, so $q_i = (200 - q_j)/2$. Symmetry gives $q^C = \frac{200}{3}$, price $\frac{200}{3}$ and $\pi^C = \left(\frac{200}{3}\right)^2 = \frac{40\,000}{9} \approx 4\,444$.

(b) Grim trigger, Cournot. Monopoly: $Q^m = 100$, $p^m = 100$, profit 10 000, so each firm produces 50 and earns $\pi^m = 5\,000$. A deviator facing $q_j = 50$ best-replies with $q_D = \frac{200-50}{2} = 75$,

price $200 - 125 = 75$ and $\pi^D = 75 \times 75 = 5\,625$. The condition is

$$\frac{5\,000}{1-\delta} \geq 5\,625 + \frac{\delta}{1-\delta} \cdot \frac{40\,000}{9} \iff \delta \geq \frac{5\,625 - 5\,000}{5\,625 - \frac{40\,000}{9}} = \frac{625}{1\,180.\overline{5}} = \frac{9}{17} \approx 0.529.$$

(c) Bertrand. With zero marginal cost and the lowest price taking the market, the unique static equilibrium is $p_1 = p_2 = 0$ with profit $\pi^N = 0$.

(d) Grim trigger, Bertrand. Each firm charges $p^m = 100$ and earns 5 000. Undercutting by ε yields (almost) the whole 10 000 in one period, and 0 afterwards:

$$\frac{5\,000}{1-\delta} \geq 10\,000 \iff \delta \geq \frac{1}{2}.$$

(e) Two-period punishment. After a deviation both firms charge 0 for two periods, then return to $p = 100$. The cooperation and deviation streams differ only in periods 0–2:

$$5\,000\,(1 + \delta + \delta^2) \geq 10\,000 \iff \delta^2 + \delta - 1 \geq 0 \iff \delta \geq \frac{\sqrt{5}-1}{2} \approx 0.618.$$

The cutoff rises from 0.5 to 0.618: the punishment costs the deviator $5\,000(\delta + \delta^2)$ rather than $5\,000\,\delta/(1-\delta)$, and the loss from a punishment that ends is smaller, so more patience is needed to deter the same one-period gain.

(d) Interpretation. The same monopoly-sharing agreement needs $\delta \geq 0.53$ under quantity competition and $\delta \geq 0.5$ under price competition. Bertrand deviation is more profitable (10 000 vs. 5 625) but its punishment is harsher (profit 0 vs. 4 444); the harsher punishment wins. Temporary punishments need more patience than permanent ones (item 13 for the quantity version).

(e) Verification. `check_T2.py` (item 9) solves the first-order conditions and recovers $q^C = \frac{200}{3}$, $\pi^C = \frac{40\,000}{9}$, $\pi^m = 5\,000$, $q_D = 75$, $\pi^D = 5\,625$, $\delta = \frac{9}{17}$ for (b), $\frac{1}{2}$ for (d), and $\frac{\sqrt{5}-1}{2} = 0.618034$ for (e). Tadelis prints no answer.

(f) Common mistakes.

- Taking the deviation profit in (b) as the monopoly profit. The deviator best-replies to the rival's 50 with 75, which is not the monopoly output.
- Forgetting in (e) that cooperation resumes: the streams coincide from period 3 on and cancel.

Answer. (a) $q^C = \frac{200}{3}$ each, $\pi^C = \frac{40\,000}{9}$. (b) $\delta \geq \frac{9}{17}$. (c) $p = 0$, zero profit. (d) $\delta \geq \frac{1}{2}$. (e) $\delta^2 + \delta \geq 1$, i.e. $\delta \geq \frac{\sqrt{5}-1}{2} \approx 0.618$: a temporary punishment is a weaker threat than a permanent one, so it requires more patience.

Item 10: Tadelis Exercise 11.2

Exam level · difficulty 3/5

Statement (verbatim from the list, item 10)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 11.2 "Holdup", p. 236 [PDF 255]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Holdup:** Consider an ultimatum game ($T = 1$ bargaining game) in which before player 1 makes his offer to player 2, player 2 can invest in the size of the pie. If player 2 chooses a low level of investment (L) then the size of the pie is small, equal to v_L, while if player 2 chooses a high level of investment (H) then the size of the pie is large, equal to v_H. The cost to player 2 of choosing L is c_L, while the cost of choosing H is c_H. Assume that $v_H > v_L > 0$, $c_H > c_L > 0$, and $v_H - c_H > v_L - c_L$.
>
> a. What is the unique subgame-perfect equilibrium of this game? Is it Pareto optimal?
>
> b. Can you find a Nash equilibrium of the game that results in an outcome that is better for both players as compared to the unique subgame-perfect equilibrium?

(a) What is asked. In an ultimatum game ($T = 1$) where player 2 first invests L or H in the size of the pie: (a) find the unique SPE and say whether it is Pareto optimal; (b) find a Nash equilibrium whose outcome is better for both players than the SPE.

(b) Method. Backward induction, then the set of Nash equilibria (Tadelis §11.1, pp. 220–224, ultimatum; holdup is the commitment problem of a sunk investment). Notation: pie v_k, cost c_k for $k \in \{L, H\}$, with $v_H > v_L$, $c_H > c_L$ and $v_H - c_H > v_L - c_L$.

(c) Solution. **(a) SPE.** In the ultimatum subgame with pie v_k, player 2 accepts any offer ≥ 0 (rejecting gives 0), so player 1 offers 0 and keeps v_k. Player 2's payoff is therefore $-c_k$ whatever she invests, and she picks the cheaper option, L (since $c_L < c_H$). The unique SPE:

- player 2 invests L;
- player 1 offers 0 after either investment; player 2 accepts every offer.

Payoffs $(v_L,\ -c_L)$. It is *not* Pareto optimal: the total surplus under H is $v_H - c_H$, larger than $v_L - c_L$ by assumption, and a transfer to player 2 can make both better off (part b). This is the holdup problem: player 2 pays the investment cost, but player 1 captures the whole pie, so she under-invests.

(b) A better Nash equilibrium. Pick a share y with

$$c_H - c_L \ < \ y \ < \ v_H - v_L,$$

which is non-empty because $v_H - c_H > v_L - c_L \iff v_H - v_L > c_H - c_L$. Strategies:

- player 2: invest H; after H accept iff her share is at least y; after L accept every offer;
- player 1: offer y after H, offer 0 after L.

Outcome: H, share y accepted, payoffs $(v_H - y,\ y - c_H)$. Check that each strategy is a best reply. Player 2 gets $y - c_H > -c_L$ from H, and deviating to L gives $-c_L$ (she then receives 0).

Player 1 gets $v_H - y > v_L$ and any other offer after H is either rejected (0) or smaller for him. The outcome beats the SPE for both players: $v_H - y > v_L$ and $y - c_H > -c_L$.

It is not subgame perfect. After H, rejecting an offer $x \in (0, y)$ gives player 2 $0 < x$: an empty threat. Its only role is to make player 1 pay y, which he does because he believes the threat.

(d) Interpretation. When the investment is sunk before a take-it-or-leave-it offer, the investor is held up. A commitment to share the surplus (a contract, a reputation for fairness, or the threat of rejection) restores efficiency, but without commitment backward induction gives the inefficient outcome.

(e) Verification. `check_T2.py` (item 10) enumerates all pure-strategy profiles on an integer grid ($v_L = 6$, $v_H = 12$, $c_L = 2$, $c_H = 5$; offers $0, \ldots, v_k$; acceptance thresholds): $(v_L, -c_L) = (6, -2)$ is a Nash outcome, and the Nash outcomes that give both players more use H with $y \in \{4, 5\}$, i.e. $3 < y < 6$. Tadelis prints no answer.

(f) Common mistakes.

- Saying that player 2 invests H because the pie is larger. In the SPE she gets nothing from it.
- Choosing y outside $(c_H - c_L,\ v_H - v_L)$: below the lower bound she prefers L, above the upper bound player 1 prefers the SPE.

Answer. (a) Unique SPE: player 2 invests L, player 1 offers 0 after either investment and player 2 accepts everything; payoffs $(v_L, -c_L)$. It is not Pareto optimal. (b) Any y with $c_H - c_L < y < v_H - v_L$ (non-empty by the assumption on net surplus): player 2 invests H and accepts offers $\geq y$, player 1 offers y after H. Payoffs $(v_H - y, y - c_H)$ beat the SPE for both, but the rejection threat is not credible.

Item 11: Gibbons Problem 2.19

Exam level · difficulty 3/5

Statement (verbatim from the list, item 11)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems (Section 2.4), Problem 2.19, p. 137 [PDF 144]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.19. In the three-period version of Rubinstein's bargaining model analyzed in Section 2.1.D, we solved for the backwards-induction outcome. What is the subgame-perfect Nash equilibrium?

Referred to in the statement. The three-period bargaining model of Section 2.1.D and its backwards-induction outcome (Gibbons pp. 68–70 [PDF p. 75–77]) (reproduced from the book):

> Players 1 and 2 are bargaining over one dollar. They alternate in making offers: first player 1 makes a proposal that player 2 can accept or reject; if 2 rejects then 2 makes a proposal that 1 can accept or reject; and so on. Once an offer has been rejected, it ceases to be binding and is irrelevant to the subsequent play of the game. Each offer takes one period, and the players are impatient: they discount payoffs received in later periods by the factor δ per period, where $0 < \delta < 1$.
>
> A more detailed description of the timing of the three-period bargaining game is as follows.
>
> (1a) At the beginning of the first period, player 1 proposes to take a share s_1 of the dollar, leaving $1 - s_1$ for player 2.
>
> (1b) Player 2 either accepts the offer (in which case the game ends and the payoffs s_1 to player 1 and $1 - s_1$ to player 2 are immediately received) or rejects the offer (in which case play continues to the second period).
>
> (2a) At the beginning of the second period, player 2 proposes that player 1 take a share s_2 of the dollar, leaving $1 - s_2$ for player 2. (Note the convention that s_t always goes to player 1, regardless of who made the offer.)
>
> (2b) Player 1 either accepts the offer (in which case the game ends and the payoffs s_2 to player 1 and $1 - s_2$ to player 2 are immediately received) or rejects the offer (in which case play continues to the third period).
>
> (3) At the beginning of the third period, player 1 receives a share s of the dollar, leaving $1 - s$ for player 2, where $0 < s < 1$.
>
> In this three-period model, the third-period settlement $(s, 1 - s)$ is given exogenously. [...]
>
> To solve for the backwards-induction outcome of this three-period game, we first compute player 2's optimal offer if the second period is reached. Player 1 can receive s in the third period by rejecting player 2's offer of s_2 this period, but the value this period of receiving s next period is only δs. Thus, player 1 will accept s_2 if and only if $s_2 \geq \delta s$. (We assume that each player will accept an offer if indifferent between accepting and rejecting.) Player 2's second-period decision problem therefore amounts to choosing between receiving $1 - \delta s$ this period (by offering $s_2 = \delta s$ to player 1) and receiving $1 - s$ next period (by offering player 1 any $s_2 < \delta s$). The discounted value of the latter option is $\delta(1 - s)$, which is less than the $1 - \delta s$ available from the former option, so player 2's optimal second-period offer is $s_2^* = \delta s$. Thus, if play reaches the second period, player 2 will offer s_2^* and player 1 will accept.
>
> Since player 1 can solve player 2's second-period problem as well as player 2 can, player 1 knows that player 2 can receive $1 - s_2^*$ in the second period by rejecting player 1's offer of s_1 this period, but the value this period of receiving $1 - s_2^*$ next period is only $\delta(1 - s_2^*)$. Thus, player 2 will accept $1 - s_1$ if and only if $1 - s_1 \geq \delta(1 - s_2^*)$, or $s_1 \leq 1 - \delta(1 - s_2^*)$. Player 1's first-period decision problem therefore amounts to choosing between receiving $1 - \delta(1 - s_2^*)$ this period (by offering $1 - s_1 = \delta(1 - s_2^*)$ to player 2) and receiving s_2^* next period (by offering any $1 - s_1 < \delta(1 - s_2^*)$ to player 2). The discounted value of the latter option is $\delta s_2^* = \delta^2 s$, which is less than the $1 - \delta(1 - s_2^*) = 1 - \delta(1 - \delta s)$ available from the former option, so player 1's optimal first-period offer is $s_1^* = 1 - \delta(1 - s_2^*) = 1 - \delta(1 - \delta s)$. Thus, in the backwards-induction outcome of this three-period game, player 1 offers the settlement $(s_1^*, 1 - s_1^*)$ to player 2, who accepts.

(a) What is asked. In the three-period Rubinstein model of Section 2.1.D, whose backward-induction outcome we solved, what is the subgame-perfect Nash equilibrium?

(b) Method. Gibbons §2.1.D (pp. 68–70). The pie is 1 and the discount factor δ. Period 1: player 1 proposes $(s_1, 1-s_1)$ and player 2 accepts or rejects. Period 2: player 2 proposes $(s_2, 1-s_2)$ and player 1 accepts or rejects. Period 3: player 1 receives s and player 2 receives $1-s$, with $0 < s < 1$. The backward-induction outcome is the *path*; the SPNE is the complete strategy of each player, that is, an action at every history, including those that never occur.

(c) Solution. Backward induction. In period 2 player 1 accepts s_2 iff $s_2 \geq \delta s$ (rejecting gives him s one period later). Player 2 proposes $s_2^* = \delta s$ and keeps $1-\delta s$. In period 1 player 2 accepts s_1 iff $1 - s_1 \geq \delta(1 - s_2^*) = \delta(1-\delta s)$. Player 1 proposes

$$s_1^* = 1 - \delta(1-\delta s) = 1 - \delta + \delta^2 s,$$

and it is accepted: payoffs $\left(1 - \delta + \delta^2 s,\ \delta - \delta^2 s\right)$.

The SPNE.

- *Player 1:* in period 1 propose $s_1^* = 1 - \delta(1-\delta s)$; in period 2, after any proposal s_2 by player 2, accept iff $s_2 \geq \delta s$.
- *Player 2:* in period 1, after any proposal s_1, accept iff $1 - s_1 \geq \delta(1-\delta s)$, i.e. iff $s_1 \leq s_1^*$; in period 2 propose $s_2^* = \delta s$.

Both strategies specify behaviour after every proposal, not only the equilibrium one. This is what makes it an SPNE rather than just a description of the outcome.

Verification (one-stage deviations). *Period 2.* If player 1 rejects $s_2 \geq \delta s$ he gets $\delta s \leq s_2$ in value. If player 2 offers $s_2 < \delta s$ it is rejected and she gets $\delta(1-s) < 1 - \delta s$ (the last inequality is $1 - \delta s - \delta + \delta s = 1 - \delta > 0$), and if she offers more than δs she gives away surplus. *Period 1.* Player 2 compares accepting $1 - s_1$ with rejecting and getting $\delta(1-\delta s)$ in value. Player 1 compares demanding $s_1 \leq s_1^*$ (accepted, payoff $s_1 \leq s_1^*$) with demanding more (rejected, payoff $\delta^2 s$). Since $s_1^* - \delta^2 s = 1 - \delta > 0$, the best demand is s_1^*. At all histories the strategies are best replies, so the profile is subgame perfect.

(d) Interpretation. The backward-induction outcome only says that agreement occurs immediately at $(s_1^*, 1 - s_1^*)$. The SPNE adds the threats and acceptance rules that make this outcome optimal, and they hold also after off-path offers. Such rules are the content of Rubinstein's argument (item 19 verifies the stationary version in the infinite horizon).

(e) Verification. `check_T2.py` (item 11) tests the formulas for three parameter pairs ($\delta = 0.9$, $s = 0.5$; $\delta = 0.5$, $s = 0.3$; $\delta = 0.8$, $s = 0.9$): $s_2 = \delta s$, $s_1 = 1 - \delta + \delta^2 s$, player 2 indifferent at s_1^*, and on a grid of step $\frac{1}{1000}$ no demand by player 1 beats s_1^* given the acceptance rule. Gibbons prints no answer.

Answer. Player 1: propose $s_1^* = 1 - \delta(1-\delta s)$ in period 1; in period 2 accept iff $s_2 \geq \delta s$. Player 2: in period 1 accept iff $1 - s_1 \geq \delta(1-\delta s)$; in period 2 propose $s_2^* = \delta s$. On the path player 2 accepts s_1^* and the payoffs are $\left(1 - \delta + \delta^2 s,\ \delta - \delta^2 s\right)$.

Item 12: Tadelis Exercise 11.5

Exam level · difficulty 4/5

Statement (verbatim from the list, item 12)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 11.5 "Asymmetric Patience 1", pp. 236–237 [PDF 255–256]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Asymmetric Patience 1:** Consider a three-period sequential (alternating-offer) bargaining model in which two players have to split a pie worth 1 (starting with player 1 making the offer). Now the players have different discount factors, δ_1 and δ_2.
>
> a. Compute the outcome of the unique subgame-perfect equilibrium.
> b. Show that when $\delta_1 = \delta_2$ player 1 has an advantage.
> c. What conditions on δ_1 and δ_2 give player 2 an advantage? Why?

(a) What is asked. In three-period alternating-offer bargaining over a pie of 1, with player 1 proposing first and discount factors $\delta_1 \neq \delta_2$: (a) the SPE outcome; (b) player 1's advantage when $\delta_1 = \delta_2$; (c) when player 2 has the advantage, and why.

(b) Method. Backward induction in finite alternating-offer bargaining (Tadelis §11.2, pp. 224–227; Gibbons §2.1.D, pp. 68–70). In the last period the proposer takes everything. In each earlier period the proposer offers the responder exactly the discounted value of what the responder would get next period, and the responder accepts at indifference (the argument of Prop. 11.2, p. 223; agreement in round one, Prop. 11.3, p. 226). Slides: Lecture 4, slides 9–10 (backward induction) and slide 17 (SPE). Lecture 5's bargaining part has no slides yet. Notation as in `Notas/L04/L04-06-finite-alternating-offers.md`.

(c) Solution. Notation. x_t = player 1's share of the pie when agreement is reached in period $t \in \{1, 2, 3\}$, so player 2 gets $1 - x_t$. Player i's payoff from agreement $(x, 1-x)$ in period t is δ_i^{t-1} times his share. Players 1, 2, 1 propose in periods 1, 2, 3, and no agreement in period 3 gives $(0, 0)$. "Advantage" means a share above $\frac{1}{2}$.

Step 1: period 3 (player 1 proposes, last period). Player 2 compares accepting $1 - x$ with rejecting, which gives 0. She accepts every $x < 1$ strictly. At $x = 1$ she is indifferent, and in an SPE she must accept: if she rejected $x = 1$, player 1 would want the largest $x < 1$, which does not exist, so he would have no best offer. Player 1 offers $x_3 = 1$. Period-3 values: player 1 gets 1, player 2 gets 0; in period-1 terms, $(\delta_1^2,\ 0)$.

Step 2: period 2 (player 2 proposes). Player 1 accepts x iff x is at least what rejecting gives him, namely 1 one period later, worth $\delta_1 \cdot 1$ today:

$$x \ \geq \ \delta_1 \, x_3 = \delta_1. \tag{1}$$

Player 2 keeps the most she can by offering exactly $x_2 = \delta_1$ (by the same argument as in Step 1, player 1 accepts at equality). She then gets $1 - \delta_1$. Making an offer player 1 rejects gives her $\delta_2 \cdot 0 = 0 < 1 - \delta_1$, so she makes the acceptable offer.

Step 3: period 1 (player 1 proposes). Player 2 accepts x iff her share now is at least the value of rejecting, i.e. $1 - x_2$ one period later, discounted by her own δ_2:

$$1 - x \ \geq \ \delta_2 \, (1 - x_2) = \delta_2 \, (1 - \delta_1).$$

Player 1 keeps as much as possible:

$$x_1 = 1 - \delta_2(1 - \delta_1) = 1 - \delta_2 + \delta_1\delta_2. \tag{2}$$

He prefers this to a rejected offer, which gives him $\delta_1 x_2 = \delta_1^2$: indeed $x_1 - \delta_1^2 = (1-\delta_1)(1+\delta_1-\delta_2) > 0$.

(a) The SPE. Strategies at every node:

- player 1: offer $x_1 = 1 - \delta_2(1 - \delta_1)$ in period 1; in period 2 accept iff offered $x \geq \delta_1$; in period 3 offer $x_3 = 1$;
- player 2: in period 1 accept iff $1 - x \geq \delta_2(1 - \delta_1)$; in period 2 offer $x_2 = \delta_1$; in period 3 accept every offer.

Each acceptance rule holds after every possible offer, including offers never made on the path, so every subgame is played optimally. The outcome is immediate agreement:

$$(v_1, v_2) = \big(1 - \delta_2(1 - \delta_1),\ \ \delta_2(1 - \delta_1)\big).$$

The tie-breaking argument of Step 1 applies at every round, so this SPE is unique.

(b) Equal discount factors. Setting $\delta_1 = \delta_2 = \delta$ in (2):

$$v_1 = 1 - \delta + \delta^2, \qquad v_2 = \delta - \delta^2,$$

$$v_1 - v_2 = 1 - 2\delta + 2\delta^2 = (1 - \delta)^2 + \delta^2 \ > \ 0.$$

Player 1 gets strictly more than half for every δ. He proposes both first and last. The edge $(1 - \delta)^2 + \delta^2$ is smallest at $\delta = \frac{1}{2}$ (derivative $-2(1 - \delta) + 2\delta = 0$), where it equals $\frac{1}{2}$ and $v_1 = \frac{3}{4}$.

(c) When player 2 is ahead. Player 2 gets more than half iff

$$\delta_2\,(1 - \delta_1) > \tfrac{1}{2} \iff \delta_1 < 1 - \frac{1}{2\delta_2}.$$

This needs $\delta_2 > \frac{1}{2}$ (otherwise $\delta_2(1 - \delta_1) \leq \frac{1}{2}$ for every δ_1) and δ_1 low enough. Example: $\delta_2 = 0.9$ gives the cut-off $\delta_1 < 1 - \frac{1}{1.8} = \frac{4}{9}$; with $\delta_1 = 0.2$, $v_2 = 0.9 \times 0.8 = 0.72$. *Why.* Player 1's power comes from the last period. An impatient player 1 values it at only δ_1 from period 2's viewpoint, so player 2 keeps $1 - \delta_1$ when she proposes. A patient player 2 values that period-2 position almost fully (δ_2), so in period 1 player 1 must concede $\delta_2(1 - \delta_1)$. Patience is bargaining power:

$$\frac{\partial v_1}{\partial \delta_1} = \delta_2 > 0, \qquad \frac{\partial v_1}{\partial \delta_2} = -(1 - \delta_1) < 0.$$

The infinite-horizon formula $v_1 = \frac{1-\delta_2}{1-\delta_1\delta_2}$ (Tadelis §11.3; `Notas/L04/L04-07`) has the same signs.

(d) Interpretation. First-mover and last-mover advantages both favour player 1, so only a large patience gap in player 2's favour can overturn them. As $\delta \to 0$ (equal patience) the first-mover advantage takes over and $v_1 \to 1$. As $\delta \to 1$ the last-mover advantage takes over and again $v_1 \to 1$, because waiting for period 3 costs player 1 nothing. Patience matters because each player's threat to reject is worth what the next round is worth *to him*: the more patient player loses less by waiting, so the other side must pay him more to settle now.

(e) Verification. `check_C_L05.py` redoes the recursion symbolically ($x_1 = 1 - \delta_2 + \delta_1\delta_2$; $v_1 - v_2 = (1 - \delta)^2 + \delta^2$ with minimum $\frac{1}{2}$ at $\delta = \frac{1}{2}$; limits 1 at both ends; the two derivatives). It tests the condition in (c) at the boundary $\delta_1 = \frac{4}{9}$ for $\delta_2 = \frac{9}{10}$ and at $\frac{4}{9} \pm \frac{1}{1000}$, confirms that player 2 can never be ahead when $\delta_2 \leq \frac{1}{2}$ (grid of step $\frac{1}{100}$), and repeats the backward induction numerically with offers on a grid of step $\frac{1}{1000}$ for three pairs; e.g. $(\delta_1, \delta_2) = (0.9, 0.7)$ gives $x_1 = 0.93$. Output: "Tadelis 11.5: v1 = 1 - d2(1 - d1), v2 = d2(1 - d1); equal delta edge (1-d)^2 + d^2 >= 1/2; P2 ahead iff d2(1-d1) > 1/2 (boundary 4/9 at d2 = 9/10 checked +-1/1000)". Tadelis prints no answer.

(f) Common mistakes.

- Discounting player 2's continuation with δ_1 (or with a common δ). Each player values the future with *his own* factor: player 2's period-1 threat is $\delta_2(1-\delta_1)$.
- Letting player 2 reject at indifference in period 3 (or player 1 in period 2). Then the proposer has no best offer and there is no SPE; acceptance at indifference is forced.
- Answering (b) with "player 1 moves first" only. With three periods he also moves last, which is why his edge does not vanish as $\delta \to 1$.

Answer. (a) Immediate agreement: player 1 gets $1-\delta_2(1-\delta_1)$, player 2 gets $\delta_2(1-\delta_1)$ (offers: $x_3 = 1$, $x_2 = \delta_1$, $x_1 = 1-\delta_2+\delta_1\delta_2$). (b) With $\delta_1 = \delta_2 = \delta$: $v_1 - v_2 = (1-\delta)^2 + \delta^2 > 0$ (at least $\frac{1}{2}$). (c) Player 2 is ahead iff $\delta_2(1-\delta_1) > \frac{1}{2}$, which needs player 2 patient ($\delta_2 > \frac{1}{2}$) and player 1 impatient ($\delta_1 < 1 - \frac{1}{2\delta_2}$). Player 1's last-period power is worth little to an impatient player 1, while player 2's period-2 power is worth a lot to a patient player 2.

Item 13: Muñoz-Garcia Ch. 6, Exercise 3

Exam level · difficulty 4/5

Statement (verbatim from the list, item 13)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 3 "Temporary Punishments from Deviation", pp. 282–283 [PDF 293–294]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider two firms competing a la Cournot, and facing linear inverse demand $p(Q) = 100 - Q$, where $Q = q_1 + q_2$ denotes aggregate output. For simplicity, assume that firms face a common marginal cost of production $c = 10$.
>
> (a) *Unrepeated game.* Find the equilibrium output each firm produces when competing a la Cournot (that is, when they simultaneously and independently choose their output levels) in the unrepeated version of the game (that is, when firms interact only once). In addition, find the profits that each firm earns in equilibrium.
>
> (b) *Repeated game—Collusion.* Assume now that the CEOs from both companies meet to discuss a collusive agreement that would increase their profits. Set up the maximization problem that firms solve when maximizing their *joint* profits (that is, the sum of profits for both firms). Find the output level that each firm should select to maximize joint profits. In addition, find the profits that each firm obtains in this collusive agreement.
>
> (c) *Repeated game—Permanent punishment.* Consider a grim-trigger strategy in which every firm starts colluding in period 1, and it keeps doing so as long as both firms colluded in the past. Otherwise, every firm deviates to the Cournot equilibrium thereafter (that is, every firm produces the Nash equilibrium of the unrepeated game found in part a forever). In words, this says that the punishment of deviating from the collusive agreement is *permanent*, since firms never return to the collusive outcome. For which discount factors this grim-trigger strategy can be sustained as the SPNE of the infinitely-repeated game?
>
> (d) *Repeated game—Temporary punishment.* Consider now a "modified" grim-trigger strategy. Like in the grim-trigger strategy of part (c), every firm starts colluding in period 1, and it keeps doing so as long as both firms colluded in the past. However, if a deviation is detected by either firm, every firm deviates to the Cournot equilibrium during only 1 period, and then every firm returns to cooperation (producing the collusive output). Intuitively, this implies that the punishment of deviating from the collusive agreement is now *temporary* (rather than permanent) since it lasts only one period. For which discount factors this "modified" grim-trigger strategy can be sustained as the SPNE of the infinitely-repeated game?
>
> (e) Consider again the temporary punishment in part (d), but assume now that it lasts for two periods. How are your results from part (d) affected? Interpret.
>
> (f) Consider again the temporary punishment in part (d), but assume now that it lasts for three periods. How are your results from part (d) affected? Interpret.

(a) What is asked. A Cournot duopoly with $p = 100 - Q$ and $c = 10$ is repeated forever. Find (a) the one-shot Cournot outcome; (b) the collusive outcome; the minimum discount factor for the collusive output under (c) a permanent Cournot punishment and (d), (e), (f) a Cournot punishment lasting one, two and three periods.

(b) Method. One-stage deviation principle with a length-T punishment (Tadelis §10.2–10.4; the grim-trigger case is $T = \infty$). The punishment phase is a static Nash equilibrium, so it is automatically optimal. Only the cooperation phase can be profitably deviated from. Notation: π^m collusive profit, π^D best deviation profit, π^C Cournot profit.

(c) Solution. **(a)** The best response is $q_i = 45 - q_j/2$, so $q^C = 30$ and $\pi^C = (40-10)\cdot 30 = 900$.

(b) Joint profit $(90-Q)Q$ is maximised at $Q = 45$, $q^m = 22.5$; price 55 and $\pi^m = 45\cdot 22.5 = 1\,012.5$.

Best deviation. Against $q_j = 22.5$, firm i produces $45 - 11.25 = 33.75$ at price 43.75, so $\pi^D = (43.75 - 10)\cdot 33.75 = 1\,139.0625 = \frac{18\,225}{16}$.

General condition for a punishment of T periods. Cooperation yields π^m in periods $0, \ldots, T$ and deviation yields π^D in period 0 and π^C in periods $1, \ldots, T$; afterwards both return to π^m and the streams cancel. No deviation pays iff

$$\sum_{j=1}^{T} \delta^j \left(\pi^m - \pi^C\right) \;\geq\; \pi^D - \pi^m \iff \sum_{j=1}^{T} \delta^j \;\geq\; \frac{126.5625}{112.5} = \frac{9}{8}.$$

(c) Permanent punishment. $T = \infty$: $\dfrac{\delta}{1-\delta} \geq \dfrac{9}{8} \iff \delta \geq \dfrac{9}{17} = \dfrac{\pi^D - \pi^m}{\pi^D - \pi^C} \approx 0.529$.

(d) One period. $\delta \geq \frac{9}{8} > 1$: impossible, so no $\delta \in (0,1)$ sustains cooperation. The loss from one Cournot period ($112.5\,\delta$) is smaller than the one-period gain (126.56) even at $\delta = 1$.

(e) Two periods. $\delta + \delta^2 \geq \frac{9}{8}$, i.e. $\delta \geq \dfrac{-1 + \sqrt{1 + 4.5}}{2} = 0.6726$.

(f) Three periods. $\delta + \delta^2 + \delta^3 \geq \frac{9}{8}$ gives $\delta \geq 0.5842$.

Pattern. ∞: 0.529; three periods: 0.584; two: 0.673; one: no δ. Longer punishments need less patience; each added period helps less, because it is discounted more.

(d) Interpretation. The punishment must be long enough for its discounted cost to cover the one-period gain from cheating. A single period of competition costs too little in this game. Temporary punishments are attractive in practice when there is some chance of a mistaken signal, because cooperation resumes, but they require patient firms or longer wars.

(e) Verification. `check_T2.py` (item 13) recomputes $\pi^C = 900$, $\pi^m = 1\,012.5$, $\pi^D = 1\,139.0625$, the ratio $\frac{9}{8}$, the roots 1.125, 0.67260, 0.58424 of $\sum \delta^j = \frac{9}{8}$ and $\frac{9}{17}$ for grim trigger, and compares truncated present values on both sides of 0.6726 (e.g. $\delta = 0.68$ cooperates, 0.67 does not). **Cross-check with the book.** The printed answers to (d), (e), (f) are the same ($\delta \geq 1.125$, 0.67, 0.58). The printed answer to (c), $\delta \geq 0.901$, does not match: the same inequality $1\,012.5 \geq 1\,139.06(1-\delta) + 900\,\delta$ gives $\delta \geq 126.56/239.06 = 0.529$, the standard Cournot cutoff $\frac{9}{17}$ (it is also the answer of item 9(b)), so we treat 0.901 as an error.

(f) Common mistakes.

- Writing the deviation stream as $\pi^D + \pi^C\,\delta/(1-\delta)$ for $T < \infty$. Cooperation resumes after T periods.
- Using π^m instead of π^D as the one-period gain: the gain is $\pi^D - \pi^m$.

Answer. (a) $q^C = 30$, $\pi^C = 900$. (b) $q^m = 22.5$, $\pi^m = 1\,012.5$ (deviation: 33.75, profit 1 139.06). (c) $\delta \geq \frac{9}{17} \approx 0.529$. (d) $\delta \geq 1.125$: impossible. (e) $\delta \geq 0.673$. (f) $\delta \geq 0.584$. The book's 0.901 in (c) is inconsistent with its own numbers.

Item 14: Gibbons Problem 2.15

Exam level · difficulty 4/5

Statement (verbatim from the list, item 14)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems, Section 2.3, Problem 2.15, p. 136 [PDF 143]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.15. Suppose there are n firms in a Cournot oligopoly. Inverse demand is given by $P(Q) = a - Q$, where $Q = q_1 + \cdots + q_n$. Consider the infinitely repeated game based on this stage game. What is the lowest value of δ such that the firms can use trigger strategies to sustain the monopoly output level in a subgame-perfect Nash equilibrium? How does the answer vary with n, and why? If δ is too small for the firms to use trigger strategies to sustain the monopoly output, what is the most-profitable symmetric subgame-perfect Nash equilibrium that can be sustained using trigger strategies?

Referred to in the statement. The Cournot stage game and the trigger strategy of Section 2.3.C (Gibbons pp. 102–103 [PDF p. 109–110]) (reproduced from the book):

> Friedman (1971) was the first to show that cooperation could be achieved in an infinitely repeated game by using trigger strategies that switch forever to the stage-game Nash equilibrium following any deviation. The original application was to collusion in a Cournot oligopoly, as follows.
>
> Recall the static Cournot game from Section 1.2.A: If the aggregate quantity on the market is $Q = q_1 + q_2$, then the market-clearing price is $P(Q) = a - Q$, assuming $Q < a$. Each firm has a marginal cost of c and no fixed costs. The firms choose quantities simultaneously. In the unique Nash equilibrium, each firm produces the quantity $(a - c)/3$, which we will call the Cournot quantity and denote by q_C. Since the equilibrium aggregate quantity, $2(a - c)/3$, exceeds the monopoly quantity, $q_m \equiv (a - c)/2$, both firms would be better off if each produced half the monopoly quantity, $q_i = q_m/2$.
>
> Consider the infinitely repeated game based on this Cournot stage game when both firms have the discount factor δ. We now compute the values of δ for which it is a subgame-perfect Nash equilibrium of this infinitely repeated game for both firms to play the following trigger strategy:
>
> > Produce half the monopoly quantity, $q_m/2$, in the first period. In the tth period, produce $q_m/2$ if both firms have produced $q_m/2$ in each of the $t-1$ previous periods; otherwise, produce the Cournot quantity, q_C.

(a) What is asked. In an infinitely repeated Cournot oligopoly with n firms and $P(Q) = a - Q$: (i) the lowest δ for which trigger strategies sustain the monopoly output; (ii) how it varies with n, and why; (iii) if δ is lower, the most profitable symmetric SPNE that trigger strategies can sustain.

(b) Method. Grim trigger with reversion to the stage Nash equilibrium (Gibbons §2.3.C, pp. 102–104). The statement gives no cost, so $c = 0$ (a positive c replaces a by $a - c$ throughout). For a symmetric output q, the profit is $\pi(q) = (a - nq)q$, the best deviation against the others' q is $q_D(q) = \frac{a-(n-1)q}{2}$ with profit q_D^2, and the punishment payoff is the Cournot profit.

(c) Solution. **Stage values.** Cournot: $q^C = \frac{a}{n+1}$, $\pi^C = \frac{a^2}{(n+1)^2}$. Monopoly output split equally: $q^m = \frac{a}{2n}$, $\pi^m = \frac{a^2}{4n}$. Deviation against q^m: $q_D = \frac{a-(n-1)a/(2n)}{2} = \frac{a(n+1)}{4n}$ and $\pi^D = \frac{a^2(n+1)^2}{16n^2}$.

(i) The cutoff. Cooperation is an SPNE iff $\frac{\pi^m}{1-\delta} \geq \pi^D + \frac{\delta\,\pi^C}{1-\delta}$, i.e. $\delta \geq \delta^* = \frac{\pi^D-\pi^m}{\pi^D-\pi^C}$. With $\pi^D - \pi^m = \frac{a^2(n-1)^2}{16n^2}$ and $\pi^D - \pi^C = \frac{a^2(n-1)^2(n^2+6n+1)}{16n^2(n+1)^2}$,

$$\delta^*(n) = \frac{(n+1)^2}{n^2+6n+1} = 1 - \frac{4n}{n^2+6n+1}.$$

For $n = 2$ this is $\frac{9}{17}$.

(ii) Variation with n. $\frac{4n}{n^2+6n+1} = \frac{4}{n+6+1/n}$ falls in n for $n \geq 1$, so δ^* *rises* with n and tends to 1 ($n = 2$: 0.529; $n = 3$: 0.571; $n = 5$: 0.643; $n = 10$: 0.752). With more firms, each firm's share of the cartel profit shrinks ($\pi^m \propto 1/n$), while the temptation to expand output stays large and the punishment (Cournot profit) is already small, so collusion needs more patience.

(iii) Below δ^*. Consider symmetric output $q \in [q^m, q^C]$ and the same trigger. No deviation pays iff $g(q) = \pi(q) - (1-\delta)q_D(q)^2 - \delta\pi^C \geq 0$, a concave quadratic in q with $g(q^C) = 0$ (at q^C deviating gains nothing). Writing it as $-(Aq^2 + Bq + C)$ with

$$A = n + \frac{(1-\delta)(n-1)^2}{4}, \quad C = a^2\Big[\frac{1-\delta}{4} + \frac{\delta}{(n+1)^2}\Big],$$

q^C is one root and the other, by the product of roots C/A, is

$$\underline{q}(\delta) = \frac{C}{A\,q^C} = \frac{(n+1)\big[\frac{1-\delta}{4} + \frac{\delta}{(n+1)^2}\big]}{n + \frac{(1-\delta)(n-1)^2}{4}}\; a.$$

Every $q \in [\underline{q}(\delta), q^C]$ is sustainable. Profit is decreasing in q above q^m, so the most profitable symmetric SPNE with trigger strategies has output $\underline{q}(\delta)$, which equals q^C at $\delta = 0$ and q^m at $\delta = \delta^*$. Example: $n = 3$, $\delta = \frac{1}{2}$: $\underline{q} = \frac{5}{28}a$, between $q^m = \frac{a}{6}$ and $q^C = \frac{a}{4}$.

(d) Interpretation. More firms make a price-fixing cartel harder (Cabral's structural factor: fewer, similar firms). Between 0 and δ^* partial collusion survives: the trigger can support output closer to the Cournot level the less patient the firms are.

(e) Verification. `check_T2.py` (item 14) derives $\delta^*(n)$ symbolically, checks $\delta^*(2) = \frac{9}{17}$, monotonicity for $n = 2, \ldots, 11$ and the limit 1. It also verifies $\underline{q}(\delta^*) = \frac{1}{2n}$ and $\underline{q}(0) = \frac{1}{n+1}$, and, for $n = 3$ and $\delta = \frac{1}{2}$, that the smallest output on a grid of step 10^{-5} satisfying the no-deviation constraint is $\frac{5}{28}$. Gibbons's unofficial manual is not copied.

(f) Common mistakes.

- Using $q_D = q^C$ as the deviation output: a deviator best-replies to q^m of the others.
- Choosing the lowest sustainable profit in (iii). Since profit falls in q above q^m, the *smallest* feasible output is the best.

Answer. (i) $\delta^* = \frac{(n+1)^2}{n^2+6n+1}$ (so $\frac{9}{17}$ for $n = 2$). (ii) It increases with n and tends to 1: more firms mean a smaller cartel share against an unchanged temptation. (iii) For $\delta < \delta^*$ the best symmetric trigger SPNE has output $\underline{q}(\delta) = \frac{(n+1)[(1-\delta)/4 + \delta/(n+1)^2]}{n+(1-\delta)(n-1)^2/4}\,a \in (q^m, q^C)$.

Item 15: Tadelis Exercise 10.4

Exam level · difficulty 3/5

Statement (verbatim from the list, item 15)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.4 "Trust Off the Equilibrium Path", p. 216 [PDF 235]. Reproduced verbatim. Figure 10.1 (p. 205) is not part of the exercise text; it is added because the statement refers to it. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Trust Off the Equilibrium Path:** Recall the trust game depicted in Figure 10.1. We argued that for $\delta \geq \frac{1}{2}$ the following pair of strategies is a subgame-perfect equilibrium. For player 1: "In period 1 I will trust player 2, and as long as there were no deviations from the pair (T, C) in any period, then I will continue to trust him. Once such a deviation occurs then I will not trust him forever after." For player 2: "In period 1 I will cooperate, and as long as there were no deviations from the pair (T, C) in any period, then I will continue to do so. Once such a deviation occurs then I will deviate forever after." Show that if instead player 2 uses the strategy "as long as player 1 trusts me I will cooperate" then the path (T, C) played forever is a Nash equilibrium for $\delta \geq \frac{1}{2}$ but is not a subgame-perfect equilibrium for any value of δ.

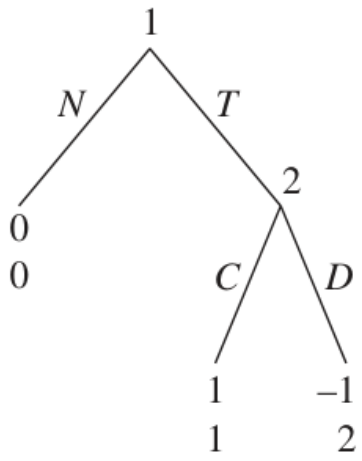


FIGURE 10.1 The one-shot trust game.

Figure 10.1 The one-shot trust game (Tadelis p. 205; added for reference).

(a) What is asked. In the repeated trust game of Figure 10.1, show that if player 2 uses "as long as player 1 trusts me I will cooperate", the path (T, C) forever is a Nash equilibrium for $\delta \geq \frac{1}{2}$ but is not subgame perfect for any δ.

(b) Method. The stage game (Figure 10.1, p. 205): player 1 chooses N (payoffs $(0, 0)$) or T; after T player 2 chooses C (payoffs $(1, 1)$) or D (payoffs $(-1, 2)$). The text's SPE (p. 205, §10.5.1) has player 1 trusting until a deviation from (T, C) and never afterwards. Player 2's *modified* strategy is σ_2: "cooperate whenever trusted", whatever the history. Nash equilibrium needs best replies on the path only. Subgame perfection needs them after every history.

(c) Solution. Let σ_1 be: "trust in period 1; as long as (T, C) was played in every earlier period keep trusting; after any other history never trust."

Nash equilibrium for $\delta \geq \frac{1}{2}$.

- *Player 1.* Against σ_2 player 2 cooperates after every trust, so trusting forever gives 1 per period and any period of N gives 0. σ_1 is a best reply.
- *Player 2.* Against σ_1, a deviation to D in some period yields 2 then no trust ever again (0), against $\frac{1}{1-\delta}$ from cooperating, which is at least 2 iff $\delta \geq \frac{1}{2}$. After the deviation σ_1 never trusts, so σ_2's prescription is never tested.

So (σ_1, σ_2) is a Nash equilibrium for $\delta \geq \frac{1}{2}$, with outcome (T, C) forever.

Not subgame perfect, for any δ. Consider a history in which player 2 defected in period t, so σ_1 prescribes no trust forever, and then the (off-path) situation in which player 1 trusts in period $t+1$. At that node player 2 chooses between C and D. Under σ_2 she cooperates and gets 1. Her continuation afterwards is 0 in both cases, because σ_1 never trusts again after the earlier defection. So

$$C: \ 1+0 \quad < \quad D: \ 2+0 \qquad \text{for every } \delta.$$

σ_2 is not a best reply in that subgame, hence the profile is not subgame perfect. The text's strategy "deviate forever after a deviation" is a best reply there, which is why it works.

(d) Interpretation. Nash equilibrium does not test choices after histories that the equilibrium itself never reaches. The text's version of player 2's strategy punishes off the path too, and therefore is subgame perfect; the version in this exercise leaves a threat (player 1's refusal to trust after a defection) that player 2 would never *need* to respect and, at the off-path node, would rather exploit.

(e) Verification. `check_T2.py` (item 15) solves $\frac{1}{1-\delta} = 2$ ($\delta = \frac{1}{2}$), checks the on-path inequality at $\delta \in \{\frac{1}{2}, \frac{3}{4}, \frac{9}{10}\}$, the strict inequality $D > C$ in the off-path subgame for $\delta \in \{\frac{1}{10}, \frac{1}{2}, \frac{99}{100}\}$, and that stopping trust at any date $k \leq 5$ is not a profitable deviation for player 1 at $\delta = 0.9$. Tadelis prints no answer.

(f) Common mistakes.

- Treating the off-path node as irrelevant because it is never reached. It is a subgame, and SPE requires best replies there.
- Concluding that the failure depends on δ: after a defection the future is zero in both branches, so the comparison $1 < 2$ holds for all δ.

Answer. With σ_2 = "cooperate whenever trusted" and σ_1 = grim trust, (T, C) forever is Nash iff $\frac{1}{1-\delta} \geq 2$, i.e. $\delta \geq \frac{1}{2}$. It is not subgame perfect for any δ: after a past defection, if player 1 trusted once, player 2 would get 1 from C against 2 from D, with continuation 0 in both cases, so cooperating is not a best reply.

Item 16: Milgrom & Roberts Ch. 8, Mathematical Exercise 5Exam level · difficulty 3/5

Statement (verbatim from the list, item 16)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Mathematical Exercise 5, p. 286 [PDF 301]. Reproduced verbatim. Answer in the source: none in the book.

> 5. Suppose instead that each time the parties interact, there is a probability $p > 0$ that they will interact again, and, correspondingly, a probability of $(1 - p)$ that this will be their last interaction. Then at any point, the probability that they will interact twice more is p^2, the probability of three more interactions is p^3, and so on. Suppose the parties are interested in the expected value of the total payoffs they receive. Thus, each seeks to maximize the sum of the payoff at the current round, plus p times the payoff that accrues if they interact again, plus p^2 times the payoff if they should interact a third time, plus p^3 times the payoff from a possible fourth interaction, and so on. In summation notation, the objective of each party i is to maximize $\Sigma p^n u_{in}$, where n runs from 0 (the current round) to infinity and u_{in} is the payoff to party i at the corresponding round. Show that so long as the payoff V to each party when trust is offered and honored is strictly larger than the payoff when trust is not offered (which we set equal to zero in the text), then for any values of the gain G to abusing trust and loss L to having trust not be honored there is a value of p that makes it Nash equilibrium behavior to offer trust and honor it at each round. What threats and punishments can be used in achieving this outcome? [Note: for any number Δ, $\sum_{n=1}^{\infty} p^n \Delta = p\Delta/(1 - p)$.]

(a) What is asked. In the trust game, each round is followed by another with probability p. Show that, whenever the payoff V of honoured trust exceeds the no-trust payoff 0, for any gain G from abusing trust and loss L from unhonoured trust there is a p that makes offering and honouring trust at every round a Nash equilibrium. What threats and punishments sustain it?

(b) Method. Stage payoffs (offeror, decision maker): offer and honour (V, V); offer and abuse $(-L, V + G)$; no offer $(0, 0)$ (pp. 261–263). The objective is $\sum_n p^n u_{in}$, so a stream of X per round is worth $X/(1 - p)$. Use the one-deviation test with the trigger "never offer (and never honour) after a breach".

(c) Solution. Strategies. Offeror: offer trust in every round as long as trust has always been honoured; after a breach never offer again. Decision maker: honour as long as trust has always been honoured; after a breach do not honour. After a breach play is the repeated static Nash equilibrium (no offer; dishonour if offered).

Decision maker. Honouring gives $V/(1 - p)$. Abusing gives $V + G$ today and then 0 (trust is never offered again). He honours iff

$$\frac{V}{1-p} \geq V + G \iff pV \geq G(1-p) \iff p \geq p^* = \frac{G}{G+V}.$$

Since $V > 0$, $p^* < 1$ and a value $p \in [p^*, 1)$ exists for *every* $G > 0$. (For $p = 1$ the game never ends and the stream is not summable; the exercise has $p < 1$.)

Offeror. On the path trust is honoured, so offering gives $V/(1-p) > 0$, the payoff from withholding for ever. He does not want to withhold, whatever L is. Off the path, after a breach, never offering is optimal because the decision maker is expected to abuse again and $-L < 0$.

Role of L. The loss L never appears in the cutoff: it matters only if trust is abused, which does not happen on the path. It makes the offeror's threat of withdrawing trust credible after a breach, because offering to someone expected to abuse costs L.

Threats and punishments. The punishment is the loss of the future gains from trade: the offeror withdraws trust for ever, so the decision maker gives up V in every future round, $\frac{pV}{1-p}$ in expected present value, for a one-round gain G. The decision maker's reputation for honesty is what makes trust worth offering.

(d) Interpretation. An uncertain end date works like a discount factor. A cooperative outcome that unravels under a known finite horizon (item 5) is sustained by a sufficiently high probability of continuing: with $V = 3$, $G = 2$ the threshold is $p^* = 0.4$; with $G = 10$ and $V = 1$ it is $p^* = \frac{10}{11}$.

(e) Verification. `check_T2.py` (item 16) solves $\frac{V}{1-p} = V + G$ symbolically ($p^* = \frac{G}{V+G}$) and checks, for $(V, G) \in \{(3, 2), (1, 10), (10, 1)\}$, that $p^* \in (0, 1)$, equality at p^* and the correct sign of the inequality at $p^* \pm 0.01$. The book prints no answer.

Answer. Honour trust iff $\frac{V}{1-p} \geq V + G$, i.e. $p \geq p^* = \frac{G}{G+V}$, which lies in $(0, 1)$ for any G, L because $V > 0$. The offeror's threat is to withdraw trust for ever after an abuse (credible, since offering to an expected abuser costs L); the decision maker's punishment is losing V in all future rounds.

Item 17: Milgrom & Roberts Ch. 8, Mathematical Exercise 3Exam level · difficulty 3/5

Statement (verbatim from the list, item 17)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Mathematical Exercise 3, p. 286 [PDF 301]. Reproduced verbatim. Answer in the source: none in the book.

> 3. In the efficiency wage model, suppose there is some probability q that the worker will appear to be cheating, even though he or she has actually worked hard and honestly. Determine the proper efficiency wage in that case.

(a) What is asked. In the efficiency-wage model, an honest, hard-working employee appears to be cheating with probability q. Determine the proper efficiency wage.

(b) Method. The model of pp. 250–255: cheating gains g once; the employee is caught with probability p and then loses the rent $w - \bar{w}$ for N periods (the multiplier N values the stream); honesty is incentive compatible iff $p(w - \bar{w})N \geq g$, giving $w = \bar{w} + g/(Np)$ (formula 8.3). The exercise adds a false alarm. I assume the evidence against a cheater is independent of the false alarm: a cheater is flagged with probability $p + (1-p)q$ (detected, or falsely flagged), an honest worker with probability q, and a flagged worker is dismissed. N is the value of one unit of per-period rent over the horizon of an employee who stays honest.

(c) Solution. Incentive constraint. Compare the expected loss of rent from cheating with that from working. Whether he cheats or not the worker is dismissed with probability q from a false alarm, so the *extra* probability of dismissal caused by cheating is

$$[p + (1-p)q] - q = p(1-q).$$

Cheating is deterred iff $p(1-q)(w - \bar{w})N \geq g$, so the minimum deterrent wage is

$$w^* = \bar{w} + \frac{g}{N\,p\,(1-q)}.$$

Compared with $q = 0$, the wage premium is multiplied by $\frac{1}{1-q}$ and rises with q: $\partial w^*/\partial q = \frac{g}{Np(1-q)^2} > 0$.

Reading. False alarms reduce the effective deterrence: a cheater was going to risk dismissal anyway (probability q), so cheating adds only $p(1-q)$ to it. The firm must pay a larger rent to restore the gap between the expected losses of cheating and honesty. If the evidence is instead read as "a cheater is flagged with probability p and an honest worker with q", the gap is $p - q$ and $w^* = \bar{w} + \frac{g}{N(p-q)}$, which also rises with q and requires $p > q$ (the monitor must be informative).

Refinement. False dismissals also shorten the effective horizon of an honest worker (it falls below the N of the no-error case), which raises the wage further. The wage formula holds for the effective N.

Numerical example. $p = \frac{1}{2}$, $g = 1\,000$, $N = 4$, $\bar{w} = 40\,000$: with $q = 0$, $w^* = 40\,500$; with $q = \frac{1}{5}$, $w^* = 40\,000 + \frac{1\,000}{4 \cdot 0.5 \cdot 0.8} = 40\,625$.

(d) Interpretation. Noisy monitoring is costly: the firm pays the employee for the risk of wrongful dismissal too. Better accuracy (smaller q) is an alternative to a higher wage, a trade-off the cost-of-monitoring problem of Mathematical Exercise 1 (item 28) puts in cost terms.

(e) Verification. `check_T2.py` (item 17) verifies $[p+(1-p)q]-q=p(1-q)$, that w^* makes the constraint bind, $w^*|_{q=0}=\bar{w}+g/(Np)$, the derivative $\frac{g}{Np(1-q)^2}$, and the example values 40 500, 40 625 and (alternative reading) $40\,000+\frac{1\,000}{4\times 0.3}$. The book prints no answer.

Answer. With false-alarm probability q and dismissal on any flag, the extra dismissal risk from cheating is $p(1-q)$, so the efficiency wage is $w^*=\bar{w}+\dfrac{g}{N\,p\,(1-q)}$, which exceeds the no-error wage $\bar{w}+g/(Np)$ by the factor $\frac{1}{1-q}$ on the premium and rises with q.

Item 18: Tadelis Exercise 10.6

Exam level · difficulty 4/5

Statement (verbatim from the list, item 18)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.6 "Regulating Medications", p. 217 [PDF 236]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Regulating Medications:** Consider a firm (player 1) that produces a unique drug that is used by a consumer (player 2). This drug is regulated by the government so that its price is $p = 6$. This price is fixed, but the quality of the drug depends on the manufacturing procedure. The "good" (G) manufacturing procedure costs 4 to the firm and yields a value of 9 to the consumer. The "bad" (B) manufacturing procedure costs 0 to the firm and yields a value of 4 to the consumer. The consumer can choose whether or not to buy at the price p, and this decision must be made before the actual manufacturing procedure is revealed. However, after consumption, the true quality of the drug is revealed to the consumer. The choice of manufacturing procedure, and thus of the cost of production, is made before the firm knows whether the consumer will buy or not.
>
> a. Draw the game tree and the matrix of this game, and find all the Nash equilibria.
> b. Now assume that the game is repeated twice. (The consumer learns the quality of the product in each period only if he consumes.) Assume that each player tries to maximize the (nondiscounted) sum of his stage payoffs. Find all the subgame-perfect equilibria of this game.
> c. Now assume that the game as repeated infinitely many times. Assume that each player tries to maximize the discounted sum of his or her stage payoffs, where the discount rate is $\delta \in (0, 1)$. What is the range of discount factors for which the good manufacturing procedure will be used as part of a subgame-perfect equilibrium?
> d. Consumer advocates are pushing for a lower price for the drug, say 5. The firm wants to approach the Food and Drug Administration and argue that if the regulated price is decreased to 5 then this may have dire consequences for both consumers and the firm. Can you make a formal argument, using the given parameters, to support the firm? What about the consumers?

(a) What is asked. A firm chooses the manufacturing procedure G (cost 4, value 9) or B (cost 0, value 4) for a drug with regulated price $p = 6$; the consumer buys or not before seeing quality. (a) The matrix and all Nash equilibria. (b) All SPE when the game is played twice. (c) The discount factors for which G is used in an SPE of the infinitely repeated game. (d) The effect of lowering the price to 5.

(b) Method. Stage game from the payoffs. The firm pays its cost whether or not the consumer buys (it chooses the procedure before knowing his decision). One-stage deviation with the trigger "revert to the stage equilibrium" (Tadelis §10.5.1, pp. 204–205; §10.3).

(c) Solution. **(a) Stage game.** Firm (rows) G, B; consumer (columns) Buy, Not. Payoffs (firm, consumer):

	Buy	Not
G	$(6-4,\ 9-6) = (2,\ 3)$	$(-4,\ 0)$
B	$(6-0,\ 4-6) = (6,\ -2)$	$(0,\ 0)$

B strictly dominates G for the firm ($6 > 2$ and $0 > -4$). The consumer's best reply to B is Not ($0 > -2$). The unique Nash equilibrium is (B, Not), in pure strategies (a mixed equilibrium puts weight 0 on G, since B is strictly dominant). Payoffs $(0, 0)$ against the cooperative $(2, 3)$.

(b) Played twice. The stage game has a unique Nash equilibrium, so backward induction gives (B, Not) in period 2 whatever happened, and the history cannot be rewarded or punished in period 1. In period 1 each player therefore plays the stage best reply. The unique SPE is (B, Not) in both periods. (That the consumer only sees quality when he buys does not matter: nothing in period 1 conditions period 2.)

(c) Infinite horizon. Strategies: the firm plays G and the consumer buys, as long as every earlier period was (G, Buy); after any deviation they play (B, Not) for ever, which is a stage equilibrium and so self-enforcing. A deviation by the firm to B is revealed after the consumer buys. The consumer's deviation (not buying) gives 0 now and 0 afterwards, below 3, so he never wants it. The firm deviates iff

$$\frac{2}{1-\delta} < 6 + 0 \iff \delta < \frac{2}{3}.$$

G is used in an SPE for $\delta \geq \frac{2}{3}$ (and not for $\delta < \frac{2}{3}$ with this trigger).

(d) Price 5. The stage game becomes G: $(1, 4)$ and $(-4, 0)$; B: $(5, -1)$ and $(0, 0)$. The unique stage equilibrium is still (B, Not), and the same trigger now needs $\frac{1}{1-\delta} \geq 5 \iff \delta \geq \frac{4}{5}$. In general, G at price p needs $\delta \geq \frac{4}{p}$ (and $p > 4$ for a positive margin).

Arguments. *Firm:* for $\delta \in [\frac{2}{3}, \frac{4}{5})$ a regulated price of 6 sustains quality G while a price of 5 does not: the firm switches to B, the consumer stops buying, and both are worse off (firm $2 \to 0$ per period, consumer $3 \to 0$). *Consumers:* if $\delta \geq \frac{4}{5}$, quality remains G and the consumer's surplus per period rises from 3 to 4, a transfer of 1 from the firm. The price cut only helps consumers if firms are patient enough; it is counterproductive when $\delta \in [\frac{2}{3}, \frac{4}{5})$.

(d) Interpretation. Reputation sustains quality only if the margin on honest behaviour is large relative to the one-period gain from cheating (p against $p - 4$). A price cap that squeezes the margin makes quality harder to sustain, but quality is a public good that disappears completely when it fails.

(e) Verification. `check_T2.py` (item 18) enumerates the pure-strategy equilibria of both matrices (only (B, Not)), checks the dominance of B, solves $\frac{p-4}{1-\delta} = p$ symbolically ($\delta = \frac{4}{p}$: $\frac{2}{3}$ at $p = 6$ and $\frac{4}{5}$ at $p = 5$), and tests $\delta = 0.7$, where $p = 6$ sustains G and $p = 5$ does not. Tadelis prints no answer.

(f) Common mistakes.

- Counting the cost 4 only when the consumer buys. The firm pays it either way, which makes B strictly dominant.
- Concluding that repetition twice allows G: a unique stage equilibrium unravels.

Answer. (a) Unique Nash equilibrium: (B, Not). (b) Unique SPE: (B, Not) in both periods. (c) G and Buy in every period is an SPE with trigger reversion iff $\delta \geq \frac{2}{3}$. (d) At $p = 5$ the cutoff becomes $\delta \geq \frac{4}{5}$, so for $\delta \in [\frac{2}{3}, \frac{4}{5})$ the cap destroys quality and both sides lose; for $\delta \geq \frac{4}{5}$ consumers gain 1 per period (surplus 4 instead of 3).

Tier 3 — Advanced

Item 19: Gibbons Problem 2.20

Advanced · difficulty 5/5

Statement (verbatim from the list, item 19)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems (Section 2.4), Problem 2.20, pp. 137–138 [PDF 144–145]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.20. Consider the following strategies in the infinite-horizon version of Rubinstein's bargaining model. (Recall the notational convention that the offer $(s, 1-s)$ means that Player 1 will get s and Player 2 will get $1-s$, independent of who made the offer.) Let $s^* = 1/(1+\delta)$. Player 1 always offers $(s^*, 1-s^*)$ and accepts an offer $(s, 1-s)$ only if $s \geq \delta s^*$. Player 2 always offers $(1-s^*, s^*)$ and accepts an offer $(s, 1-s)$ only if $1-s \geq \delta s^*$. Show that these strategies are a Nash equilibrium. Show that this equilibrium is subgame-perfect.

(a) What is asked. Show that the stationary strategies given in the problem (player 1 offers $(s^*, 1-s^*)$ and accepts iff $s \geq \delta s^*$; player 2 offers $(1-s^*, s^*)$ and accepts iff $1-s \geq \delta s^*$, with $s^* = 1/(1+\delta)$) are first a Nash equilibrium and then a subgame-perfect equilibrium of the infinite-horizon Rubinstein game.

(b) Method. Nash equilibrium: bound each player's payoff over every possible agreement date against the opponent's fixed strategy. Subgame perfection: the one-stage deviation principle (Tadelis Thm. 9.1, p. 185; its infinite-horizon form, Prop. 10.3, p. 198). It applies because payoffs are discounted and bounded, so a deviation's effect far in the future is negligible. Because the strategies ignore history, there are only four kinds of subgame to check. Model: Gibbons §2.1.D, pp. 68–71 (the infinite-horizon value $1/(1+\delta)$ and the Shaked–Sutton argument, pp. 70–71); Tadelis §11.3, pp. 228–229 (eqs. (11.4)–(11.5)). Slides: Lecture 4, slide 17 (SPE: Nash equilibrium in *every* subgame) and slides 9–10 (backward induction). Notation as in `Notas/L04/L04-07-rubinstein-infinite-horizon.md`.

(c) Solution. Notation and two identities. s is always player 1's share. In period t an agreement $(s, 1-s)$ is worth $\delta^{t-1}s$ to player 1 and $\delta^{t-1}(1-s)$ to player 2; perpetual disagreement gives $(0,0)$. Player 1 proposes in odd periods, player 2 in even ones. With $s^* = \frac{1}{1+\delta}$,

$$1 - s^* = \frac{\delta}{1+\delta} = \delta s^*, \qquad 1 - \delta s^* = \frac{1}{1+\delta} = s^*. \tag{3}$$

So player 2's offer $(1-s^*, s^*)$ gives player 1 exactly δs^*, his acceptance threshold, and player 1's offer gives player 2 exactly $1-s^* = \delta s^*$, her threshold. On the path, player 1 offers $(s^*, 1-s^*)$ in period 1 and player 2 accepts: payoffs $(s^*, 1-s^*) = (\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$.

Part 1: Nash equilibrium. *Player 1's best response.* Fix player 2's strategy. Whatever player 1 does, any agreement happens either in an odd period t, on an offer of player 1 that player 2 accepts, so $1-s \geq \delta s^*$, i.e. $s \leq 1-\delta s^* = s^*$ by (3); or in an even period t, on player 2's only offer, $s = 1-s^* = \delta s^*$. Player 1's payoff is therefore at most

$$\max\Big\{ \max_{t \text{ odd}} \delta^{t-1} s^*, \quad \max_{t \text{ even}} \delta^{t-1}\, \delta s^*, \quad 0 \Big\} \;=\; \max\{s^*,\ \delta^2 s^*,\ 0\} = s^*,$$

reached at $t = 1$, and his strategy gets exactly s^*. It is a best response.

Player 2's best response. Fix player 1's strategy. Agreement in an odd period t is on player 1's only offer, giving player 2 $\delta^{t-1}(1 - s^*)$. Agreement in an even period t needs an offer player 1 accepts, $s \geq \delta s^*$, so player 2 gets at most $\delta^{t-1}(1 - \delta s^*) = \delta^{t-1} s^*$. The best she can do is

$$\max\{\, 1 - s^* \ (t = 1), \ \ \delta s^* \ (t = 2), \text{ smaller values later} \,\} = 1 - s^* = \delta s^*,$$

the two candidates being equal by (3). Her strategy (accept in period 1) gets $1 - s^*$, so it is a best response. She is indifferent between accepting now and rejecting to propose next period, which is the property that pins down s^*. So the strategies form a Nash equilibrium.

Part 2: subgame perfection. The strategies do not depend on history, so a subgame is characterized only by whose move it is and, for a responder, by the offer on the table. Up to the common factor δ^{t-1} there are four types:

(i) *player 1 about to propose* (odd period);

(ii) *player 2 responding to an arbitrary offer* $(s, 1 - s)$ (odd period);

(iii) *player 2 about to propose* (even period);

(iv) *player 1 responding to an arbitrary offer* $(s, 1 - s)$ (even period).

Continuation values. Starting a type-(i) subgame, the strategies give agreement at once on $(s^*, 1 - s^*)$. Starting a type-(iii) subgame, they give agreement at once on $(1 - s^*, s^*)$: player 2 gets s^* and player 1 gets $1 - s^* = \delta s^*$. By the one-stage deviation principle it is enough to check single deviations at each type.

(ii) Player 2 responding to $(s, 1 - s)$. Accepting gives $1 - s$. Rejecting leads to a type-(iii) subgame next period, where she gets s^*, worth δs^* now. Accepting is optimal iff $1 - s \geq \delta s^*$, which is exactly her rule (at equality she is indifferent and accepting is a best reply). No one-stage deviation pays, for *every* offer s, on or off the path.

(iv) Player 1 responding to $(s, 1 - s)$. Accepting gives s. Rejecting leads to a type-(i) subgame next period, worth δs^*. Accepting is optimal iff $s \geq \delta s^*$: his rule.

(i) Player 1 proposing. If he offers $s \leq s^*$, player 2 accepts ($1 - s \geq 1 - s^* = \delta s^*$) and he gets $s \leq s^*$. If he offers $s > s^*$, player 2 rejects; next period player 2 offers $(\delta s^*, \cdot)$, he accepts, and he gets $\delta \cdot \delta s^* = \delta^2 s^* < s^*$. The best offer is $s = s^*$, as prescribed.

(iii) Player 2 proposing. Symmetrically, offering player 1 $s \geq \delta s^*$ is accepted and leaves her $1 - s \leq 1 - \delta s^* = s^*$; offering less is rejected; player 1 then proposes $(s^*, 1 - s^*)$, she accepts, and she gets $\delta(1 - s^*) = \delta^2 s^* < s^*$. Her best offer is $s = 1 - s^* = \delta s^*$, as prescribed.

No player gains from a single deviation in any subgame, so by the one-stage deviation principle the strategy profile is a Nash equilibrium in every subgame: it is subgame perfect.

Why s^* and not another number. The two indifference conditions are $1 - s_1 = \delta(1 - s_2)$ (player 2 indifferent in odd periods) and $s_2 = \delta s_1$ (player 1 indifferent in even periods). Substituting, $s_1 = 1 - \delta(1 - \delta s_1)$, whose only solution is $s_1 = \frac{1}{1+\delta} = s^*$. The same fixed point appears in Gibbons's backward-induction argument, pp. 70–71. The problem does not ask for uniqueness; Rubinstein (1982), via the Shaked–Sutton argument in Gibbons, shows this is the *only* SPE.

(d) Interpretation. With an infinite horizon there is no last period to anchor backward induction, but the game looks the same every time a given player proposes. Each proposer offers the responder exactly what the responder could get by waiting one period and proposing himself, δs^*, and keeps the rest. The first proposer's advantage is $s^* - (1 - s^*) = \frac{1-\delta}{1+\delta}$, the cost of one period of delay, and it vanishes as $\delta \to 1$, where the split tends to $(\frac{1}{2}, \frac{1}{2})$. Off the path the strategies

tell each responder to accept anything at least as good as the value of waiting: these threats are credible, which is exactly what subgame perfection requires beyond Nash.

(e) Verification. `check_C_L05.py` verifies the identities (3) and the unique solution $s^* = 1/(1+\delta)$ of $s = 1-\delta(1-\delta s)$ with sympy. Then, for $\delta \in \{\frac{1}{2}, \frac{3}{4}, \frac{9}{10}, \frac{99}{100}\}$ in exact fractions, it checks: each responder accepts at the threshold and switches exactly at threshold $\pm\frac{1}{1000}$; the proposers' best one-stage offers, on a grid of step $\frac{1}{1000}$ plus $s^* \pm \frac{1}{1000}$, are the prescribed ones, and a rejected offer yields $\delta^2 s^*$; and the best agreement date against the opponent's strategy is $t = 1$ for player 1 and $t = 1$ (tied with $t = 2$) for player 2, over 59 periods. Output: "Gibbons 2.20: s* = 1/(1+delta), 1 - s* = delta s*; one-stage deviations by proposers and responders unprofitable for delta in {1/2, 3/4, 9/10, 99/100}; cutoffs +-1/1000 pass". Gibbons prints no answer.

(f) Common mistakes.

- Proving only the Nash part, or checking only on-path behaviour. Subgame perfection needs the responder's rule to be optimal after *every* offer, including offers never made.
- Valuing a rejection by the responder's share of the *current* offer, or forgetting to discount: rejecting today is worth δs^* (the proposer's share next period, one period later), not s^*.
- Applying the one-stage deviation principle without saying why it holds in an infinite game: it needs discounting with bounded payoffs (continuity at infinity).

Answer. With $s^* = \frac{1}{1+\delta}$ we have $1 - s^* = \delta s^*$ and $1 - \delta s^* = s^*$. Nash: against player 2's strategy, player 1 can get at most s^* (odd periods) or $\delta^2 s^*$ (even periods), and his strategy gets s^*; against player 1's strategy, player 2 can get at most $1 - s^* = \delta s^*$, and accepting in period 1 gets it. SPE: in each of the four subgame types (proposer 1, responder 2, proposer 2, responder 1), no one-stage deviation pays, because each responder accepts exactly when the offer is worth at least waiting (δs^*) and each proposer's best offer is the prescribed one (a rejected offer gives $\delta^2 s^* < s^*$). By the one-stage deviation principle the profile is subgame perfect, with immediate agreement on $(\frac{1}{1+\delta}, \frac{\delta}{1+\delta})$.

Item 20: Muñoz-Garcia Ch. 6, Exercise 5

Advanced · difficulty 4/5

Statement (verbatim from the list, item 20)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 5 "Collusion When N Firms Compete in Prices", pp. 290–291 [PDF 301–302]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Consider a homogenous industry where n firms produce at zero cost and play the Bertrand game of price competition for an infinite number of periods. Assume that:
>
> - When firms choose the same price, they earn a per-period profit $\pi(p) = p\alpha\frac{D(p)}{n}$, where parameter α represents the state of demand.
> - When a firm i charges a price of p_i lower than the price of all of the other firms, it earns a profit $\pi(p_i) = p_i\alpha D(p_i)$, and all of the other firms obtain zero profits.
>
> Imagine that in the current period demand is characterized by $\alpha = 1$, but starting from the following period demand will be characterized by $\alpha = \theta$ in each of the following periods. All the players know exactly the evolution of the demand state at the beginning of the game, and firms have the same common discount factor, δ.
>
> Assume that $\theta > 1$ and consider the following trigger strategies. Each firm plays the monopoly price p_m in the first period of the game, and continues to charge such a price until a profit equal to zero is observed. When this occurs, each firm charges a price equal to zero forever. Under which conditions does this trigger strategy represent a SPNE? [Hint: In particular, show how θ and n affects such a condition, and give an economic intuition for this result.]

(a) What is asked. n Bertrand firms with zero cost; demand is $\alpha = 1$ in the current period and $\alpha = \theta > 1$ in every later period, known to all. Under the trigger "charge the monopoly price p_m until a zero profit is observed, then charge 0 for ever", find the condition for an SPNE and show how θ and n affect it.

(b) Method. One-stage deviation with a zero-profit punishment (Cabral §9.1 and §9.3, pp. 218–222, 227–231). Let $\pi(p_m)$ be the monopoly profit at $\alpha = 1$; at $\alpha = \theta$ it is $\theta\,\pi(p_m)$. Reversion to $p = 0$ forever is a stage Nash equilibrium (profit 0), so the punishment phase needs no check.

(c) Solution. Cooperation. Each firm gets $\frac{1}{n}\pi(p_m)$ now and $\frac{\theta}{n}\pi(p_m)$ in each later period:

$$V^{coop} = \frac{\pi(p_m)}{n}\Big(1 + \delta\theta + \delta^2\theta + \cdots\Big) = \frac{\pi(p_m)}{n}\Big(1 + \frac{\delta\theta}{1-\delta}\Big).$$

Deviation. Undercutting slightly wins the whole market: $\pi(p_m)$ today and 0 afterwards.

Condition.

$$\frac{1}{n}\Big(1 + \frac{\delta\theta}{1-\delta}\Big) \geq 1 \iff 1 - \delta + \delta\theta \geq n(1-\delta) \iff \delta \geq \tilde{\delta}(n,\theta) = \frac{n-1}{n-1+\theta}.$$

Comparative statics.

$$\frac{\partial\tilde{\delta}}{\partial\theta} = -\frac{n-1}{(n-1+\theta)^2} < 0, \qquad \frac{\partial\tilde{\delta}}{\partial n} = \frac{\theta}{(n-1+\theta)^2} > 0.$$

Check at $\theta = 1$ (constant demand): $\tilde{\delta} = \frac{n-1}{n} = 1 - \frac{1}{n}$, the usual n-firm cutoff; for $n = 2$ it is $\frac{1}{2}$. For $n = 3$, $\theta = 2$ it is $\frac{1}{2}$ again, and as $\theta \to \infty$ it tends to 0.

(d) Interpretation. *Growth helps collusion.* The higher θ, the more profit a firm forfeits by triggering a war, so the threat is costlier and collusion is easier ($\tilde{\delta}$ falls). *More firms hinder collusion.* The cartel share $\frac{1}{n}$ shrinks while the deviation prize (the whole market) does not ($\tilde{\delta}$ rises with n, Cabral's "few firms" factor). This is the demand-growth entry of Cabral's list of factors (the $(1+g)$ in his (9.5)).

(e) Verification. `check_T3.py` (item 20) solves the indifference condition symbolically ($\tilde{\delta} = \frac{n-1}{n-1+\theta}$), both derivatives, the case $\theta = 1$ and the limit 0 as $\theta \to \infty$, and the stream comparison for $n = 3$, $\theta = 2$ at $\delta = 0.49, 0.5, 0.51$. This matches the book's answer, which states the same cutoff and the same two signs.

(f) Common mistakes.

- Applying θ in the current period. Demand is 1 today and θ from tomorrow on.
- Forgetting that the deviation profit is the full $\pi(p_m)$, not $\frac{1}{n}\pi(p_m)$: the firm takes all demand today.

Answer. The trigger is an SPNE iff $\delta \geq \frac{n-1}{n-1+\theta}$. The cutoff falls with θ (growing demand raises the cost of a price war) and rises with n (a smaller cartel share against an unchanged deviation gain); at $\theta = 1$ it is $1 - \frac{1}{n}$.

Item 21: Cabral Exercise 9.15

Advanced · difficulty 4/5

Statement (verbatim from the list, item 21)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 9 Review and practice exercises, Exercise 9.15 "Multimarket contact", p. 244 [PDF 256]. Reproduced verbatim. Answer in the source: none in the book.

> **9.15. Multimarket contact.** Consider the model of multimarket contact presented in Section 9.3: Firm 1 has cost $\underline{c}$ in Market 1, while Firm 2 has a cost $\overline{c}$. The situation is reversed in Market 2. Demand is the same in both markets: consumers are willing to buy q units for a price of up to p^M (that is, for $p \leq p^M$, quantity demanded does not depend on price). It is assumed that $\underline{c} < \overline{c} < p^M$. In each period, firms set prices in both markets simultaneously.
>
> Determine the minimum value of the discount factor such that the optimal collusive solution is stable.

Referred to in the statement. The model of multimarket contact in Section 9.3 (Cabral pp. 228–229 [PDF p. 240–241]) (reproduced from the book; the side note it cites is kept):

> Consider a price-setting homogeneous-product duopoly where marginal cost is constant and the same for both firms. If firms set prices at the monopoly level, p^M, then each receives $\frac{1}{2}\pi^M$, where π^M is monopoly profits. If one of the firms were to set a slightly lower price, then it would receive all of the monopoly profits, that is, twice as much. The worse price war to punish such deviation would consist of setting price at marginal cost forever, that is, zero profits. [...]
>
> Now suppose that each firm has a cost advantage in one of the markets. For example, in Market 1 Firm 1's cost is $\underline{c}$, whereas Firm 2's is $\overline{c} > \underline{c}$; and in Market 2 Firm 2's cost is $\underline{c}$, whereas Firm 1's is $\overline{c}$. One possible interpretation is that each firm is located in a different country and $\overline{c} = \underline{c} + t$, where t is the transportation cost from one country to the other.
>
> As seen in the previous subsection, the efficient collusive agreement in Market 1 is for Firm 1 to set its monopoly price and for Firm 2 to set a higher price and sell zero. The reverse is true in Market 2. Taken in isolation, such agreements are clearly not stable. However, if we consider the two markets together, then the situation is quite different. Firm 1 might be able to convince Firm 2 not to undercut its monopoly price in Market 1 with the threat that it would undercut Firm 2 in the latter's market if that were to happen. In fact, suppose that the punishment from deviation would be for firms to engage in a price war whereby price equals the high marginal cost, $\overline{c}$.[j] One can easily show that if the discount factor is sufficiently high, then firms will be able to collude and set prices at the monopoly level (see Exercise 9.18). Multimarket contact matters.
>
> *j.* It can be shown that this constitutes an equilibrium of the static price game.

(a) What is asked. Two markets with the same inelastic demand (q units up to a reservation price p^M). Firm 1's cost is $\underline{c}$ in Market 1 and $\overline{c}$ in Market 2, and the reverse for Firm 2, with $\underline{c} < \overline{c} < p^M$. Firms set prices in both markets each period. Find the minimum discount factor for which the optimal collusive solution is stable.

(b) Method. Cabral §9.3 (pp. 227–231, multimarket contact). The efficient collusive agreement in each market is that the low-cost firm sets p^M and the high-cost firm sets a higher price and sells nothing. The punishment is a price war at the static equilibrium, in which the low-cost firm prices just below the rival's cost $\overline{c}$.

(c) Solution. Collusion. Each firm sells all q units at p^M in its low-cost market: profit per period $A = q(p^M - \underline{c})$.

Deviation. Take Firm 1. In Market 1 it is already a monopolist, and in Market 2 it can undercut Firm 2's price p^M slightly, gaining the whole market at cost $\overline{c}$: extra profit $B = q(p^M - \overline{c}) > 0$. So the deviation profit in the period is $A + B$.

Punishment. Static equilibrium in each market: the low-cost firm charges just below $\overline{c}$ and the high-cost firm sells nothing. Each firm earns, in its low-cost market, $q(\overline{c} - \underline{c}) = A - B$ per period. Both markets are in the war.

Condition.

$$\frac{A}{1-\delta} \geq A + B + \frac{\delta(A-B)}{1-\delta} \iff A \geq (A+B)(1-\delta) + \delta(A-B) \iff 0 \geq B - 2\delta B \iff \delta \geq \frac{1}{2}.$$

So the minimum discount factor is $\delta = \frac{1}{2}$, whatever $\underline{c}$, $\overline{c}$, p^M or q (with $\overline{c} < p^M$ so that $B > 0$).

(d) Interpretation. In a single market the efficient agreement is never stable: the high-cost firm earns zero and would undercut at any δ, since no punishment can hurt it. With two markets each firm is the "low-cost" firm in one market and has something to lose in the other, so the threat of retaliation in the rival's market holds the agreement together. This is Cabral's point that multimarket contact matters when markets are asymmetric (his airlines and dog food boxes, pp. 229–230).

(e) Verification. `check_T3.py` (item 21) solves the indifference equation symbolically with A, B and $A - B$ in terms of p^M, $\underline{c}$, $\overline{c}$, q and obtains $\frac{1}{2}$. It tests numbers ($p^M = 10$, $\underline{c} = 2$, $\overline{c} = 6$, $q = 1$: $A = 8$, $B = 4$, war profit 4) at $\delta = \frac{1}{2}, \frac{2}{3}$ (stable) and 0.49 (not stable). Cabral prints no answer.

(f) Common mistakes.

- Using zero as the punishment payoff. In the static equilibrium of this asymmetric game the low-cost firm still earns $q(\overline{c} - \underline{c})$ in its own market.
- Counting only one market's deviation gain: Firm 1 deviates in Market 2 (the rival's market) and continues to collect A in Market 1 in the same period.

Answer. $\delta \geq \dfrac{1}{2}$: with collusive profit A, extra deviation profit B and war profit $A - B$ the condition $\frac{A}{1-\delta} \geq A + B + \frac{\delta(A-B)}{1-\delta}$ reduces to $\delta \geq \frac{1}{2}$, independent of the costs, q and p^M.

Item 22: Muñoz-Garcia Ch. 6, Exercise 10

Advanced · difficulty 5/5

Statement (verbatim from the list, item 22)

Source. Muñoz-Garcia & Toro-Gonzalez (2019), *Strategy and Game Theory: Practice Exercises with Answers*, Ch. 6, Exercise 10 "Collusion and Imperfect Monitoring", p. 310 [PDF 321]. Reproduced verbatim (statement only; the book's worked answer follows it).

> Two firms produce a differentiated product, and every firm $i = \{1, 2\}$ faces the following linear demand:
>
> $$p_i = a - bq_i - dq_j, \quad \text{where } a, b, d > 0 \text{ and } 0 < d < b$$
>
> Intuitively, when $d \to 0$ firm j's output decisions do not affect firm i's sales (i.e., products are heterogeneous), while when $d \to b$, firm j's output decisions have a significant impact on firm i's sales, i.e., products are relatively homogeneous. The firms expect to compete against each other for infinite periods, that is, there is no entry or exit in the industry. [*Hint*: Assume that when a firm detects deviations from collusive behavior, it retaliates by reverting to the non-cooperative Cournot (Bertrand) production level (pricing, respectively), in all subsequent periods. Prior to detection, each firm assumes that its rival follows the collusive outcome.]
>
> **Part (a)** Assume that firms compete as Cournot oligopolists. What is the minimum discount factor necessary in order to sustain the collusive outcome assuming that (i) Deviations are detected immediately, and (ii) Deviations are detected with a lag of one period.
>
> **Part (b)** Repeat part (a) of the exercise under the assumption that firms compete as Bertrand oligopolists. (You can assume immediate detection)

(a) What is asked. Two firms sell differentiated products with $p_i = a - bq_i - dq_j$ ($0 < d < b$, zero cost), repeated for ever, with reversion to the static equilibrium after a detected deviation. (a) Under Cournot competition, find the minimum discount factor that sustains collusion when (i) deviations are detected immediately and (ii) they are detected with a one-period lag. (b) The same under Bertrand competition (immediate detection).

(b) Method. One-stage deviation: $\dfrac{\pi^m}{1-\delta} \geq$ (deviation stream). Collusion is the joint-profit maximum with each firm earning half. The punishment is the repeated static equilibrium, so only the cooperation phase needs checking. With immediate detection $\delta \geq \dfrac{\pi^D - \pi^m}{\pi^D - \pi^N}$.

(c) Solution. (a) Cournot.

- *Static.* The best reply is $q_i = (a - dq_j)/(2b)$, so $q^C = \frac{a}{2b+d}$ and $\pi^C = \frac{ba^2}{(2b+d)^2}$.
- *Collusion.* Symmetric joint profit $2\,(a - (b+d)q)\,q$ is maximised at $q^m = \frac{a}{2(b+d)}$ per firm, price $\frac{a}{2}$, so $\pi^m = \frac{a^2}{4(b+d)}$ (each firm's half of the monopoly profit).
- *Deviation.* The best reply to q^m is $q^D = \frac{a-dq^m}{2b} = \frac{a(2b+d)}{4b(b+d)}$ with profit $\pi^D = b(q^D)^2 = \frac{a^2(2b+d)^2}{16b(b+d)^2}$.

(i) Immediate detection.

$$\delta_1 = \frac{\pi^D - \pi^m}{\pi^D - \pi^C} = \frac{(2b+d)^2}{8b^2 + 8bd + d^2}.$$

At $d \to 0$ it is $\frac{1}{2}$ (independent monopolists), at $d = b$ it is $\frac{9}{17}$ (homogeneous goods), and $\frac{\partial \delta_1}{\partial d} = \frac{4bd(2b+d)}{(8b^2+8bd+d^2)^2} > 0$: collusion is harder when products are closer substitutes.

(ii) One-period lag. The deviator earns π^D in the deviation period *and* the next (the rival has not yet reacted), then π^C:

$$\frac{\pi^m}{1-\delta} \geq \pi^D(1+\delta) + \frac{\delta^2 \pi^C}{1-\delta} \iff \delta^2 \geq \frac{\pi^D - \pi^m}{\pi^D - \pi^C} = \delta_1,$$

so

$$\delta_2 = \sqrt{\delta_1} = \frac{2b+d}{\sqrt{8b^2+8bd+d^2}} > \delta_1.$$

Collusion needs more patience when detection lags.

(b) Bertrand. Inverting demand, $q_i = \dfrac{a(b-d) - bp_i + dp_j}{b^2 - d^2}$.

- *Static.* The best reply is $p_i = \frac{a(b-d)+dp_j}{2b}$, so $p^B = \frac{a(b-d)}{2b-d}$ and $\pi^B = \frac{a^2 b(b-d)^2}{(2b-d)^2(b^2-d^2)}$.
- *Collusion.* A common price maximising $2pq(p,p)$ with $q = \frac{a-p}{b+d}$: $p^m = \frac{a}{2}$ and $\pi^m = \frac{a^2}{4(b+d)}$ (the same joint outcome as under Cournot).
- *Deviation.* The best reply to $\frac{a}{2}$ is $p^D = \frac{a(2b-d)}{4b}$ with $\pi^D = \frac{a^2(2b-d)^2}{16b(b^2-d^2)}$.

$$\delta_B = \frac{\pi^D - \pi^m}{\pi^D - \pi^B} = \frac{(2b-d)^2}{8b^2 - 8bd + d^2}.$$

It tends to $\frac{1}{2}$ as $d \to 0$ and to 1 as $d \to b$ (the deviation profit explodes when products become perfect substitutes: a small price cut captures the market).

(d) Interpretation. Product differentiation helps collusion under Cournot (the deviation gain shrinks as $d \to 0$) and hurts it under Bertrand with these demands. A detection lag raises the required patience, since the deviator enjoys the gain for an extra period before the punishment begins.

(e) Verification. `check_T3.py` (item 22) derives every profit above symbolically and the three cutoffs, checks that $\delta_2^2 = \delta_1$ by solving the lag condition for δ, the derivative of δ_1, the limits $\frac{1}{2}$ and $\frac{9}{17}$, and (numerically, $a = b = 1$, $d = \frac{1}{2}$) the best replies on a grid of step 10^{-4}. **Cross-check with the book.** The legible parts of the printed answers for δ_1 and δ_2 agree with ours (numerator $4b^2 + 4bd + d^2 = (2b+d)^2$; δ_2 is a square root). Its intermediate Cournot monopoly profit as shown in the text layer (the collusive $\frac{a^2}{8(b+d)}$) differs from our $\frac{a^2}{4(b+d)}$, which we checked directly, and the Bertrand cutoff as extracted carries an extra factor $(2b^2 - 2bd + d^2)/a^2$: at $a = b = 1$, $d = \frac{1}{2}$ it gives 0.662, whereas the indifference condition holds exactly at 0.529.

(f) Common mistakes.

- Using $q^m = \frac{a}{4(b+d)}$ per firm: the joint maximum is $\frac{a}{2(b+d)}$ *each* when demands are $p_i = a - bq_i - dq_j$.
- In (ii) giving the deviator π^D for one period only. With a one-period lag it earns π^D twice.

Answer. (a)(i) $\delta \geq \frac{(2b+d)^2}{8b^2+8bd+d^2}$ (from $\frac{1}{2}$ at $d \to 0$ to $\frac{9}{17}$ at $d = b$); (ii) with a one-period lag $\delta \geq \sqrt{\cdot}$ of that, $\frac{2b+d}{\sqrt{8b^2+8bd+d^2}}$. (b) Bertrand, immediate detection: $\delta \geq \frac{(2b-d)^2}{8b^2-8bd+d^2}$.

Item 23: Gibbons Problem 2.14

Advanced · difficulty 5/5

Statement (verbatim from the list, item 23)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems, Section 2.3, Problem 2.14, pp. 135–136 [PDF 142–143]. Reproduced verbatim (the book's spelling "Rotemburg" kept). Answer in the source: none (Gibbons prints no answers).

> 2.14. Suppose that demand fluctuates randomly in the infinitely repeated Bertrand game described in Problem 2.13: in each period, the demand intercept is a_H with probability π and a_L ($< a_H$) with probability $1-\pi$; demands in different periods are independent. Suppose that each period the level of demand is revealed to both firms before they choose their prices for that period. What are the monopoly price levels (p_H and p_L) for the two levels of demand? Solve for δ^*, the lowest value of δ such that the firms can use trigger strategies to sustain these monopoly price levels (i.e., to play p_i when demand is a_i, for $i = H, L$) in a subgame-perfect Nash equilibrium. For each value of δ between $1/2$ and δ^*, find the highest price $p(\delta)$ such that the firms can use trigger strategies to sustain the price $p(\delta)$ when demand is high and the price p_L when demand is low in a subgame-perfect Nash equilibrium. (See Rotemburg and Saloner 1986.)

Referred to in the statement. Problem 2.13 (Gibbons §2.3 Problems, p. 135 [PDF p. 142]) (reproduced from the book):

> 2.13. Recall the static Bertrand duopoly model (with homogeneous products) from Problem 1.7: the firms name prices simultaneously; demand for firm i's product is $a - p_i$ if $p_i < p_j$, is 0 if $p_i > p_j$, and is $(a - p_i)/2$ if $p_i = p_j$; marginal costs are $c < a$. Consider the infinitely repeated game based on this stage game. Show that the firms can use trigger strategies (that switch forever to the stage-game Nash equilibrium after any deviation) to sustain the monopoly price level in a subgame-perfect Nash equilibrium if and only if $\delta \geq 1/2$.

(a) What is asked. In the infinitely repeated Bertrand game of Problem 2.13, demand $a - p$ has intercept a_H with probability π and $a_L < a_H$ with probability $1-\pi$, i.i.d., and the state is revealed before prices are set. Find (i) the monopoly prices p_H, p_L; (ii) the lowest δ^* for which trigger strategies sustain them; (iii) for $\delta \in [\frac{1}{2}, \delta^*)$, the highest price $p(\delta)$ sustainable in the high state with p_L in the low state.

(b) Method. One-stage deviation with reversion to the static equilibrium (price $= c$, profit 0), Gibbons §2.3.C and the Rotemberg–Saloner model of the text. In state i, each firm's collusive profit is half the monopoly profit and a price-cutter takes all of it. Let $\pi_i = \frac{(a_i - c)^2}{8}$ (each firm's collusive profit) and $\bar{\pi} = \pi\,\pi_H + (1-\pi)\,\pi_L$ (expected, before the state is drawn).

(c) Solution. (i) Monopoly prices. Maximising $(p-c)(a_i - p)$: $p_i = \frac{a_i + c}{2}$, with total profit $\frac{(a_i-c)^2}{4}$ and $\pi_i = \frac{(a_i-c)^2}{8}$ per firm.

(ii) The cutoff δ^*. In state i, collusion gives $\pi_i + \frac{\delta}{1-\delta}\bar{\pi}$ (today's profit plus the expected value of the future, which is the same whether or not today's state is high) and deviation gives $2\pi_i$ (then 0). The condition is

$$\frac{\delta}{1-\delta}\,\bar{\pi} \geq \pi_i \iff \delta \geq \frac{\pi_i}{\pi_i + \bar{\pi}}.$$

It is hardest in the high state (larger π_i), so

$$\delta^* = \frac{\pi_H}{\pi_H + \bar{\pi}} = \frac{(a_H - c)^2}{(a_H - c)^2(1+\pi) + (1-\pi)(a_L - c)^2} \in \left(\tfrac{1}{2}, \tfrac{1}{1+\pi}\right).$$

Example: $a_H = 10$, $a_L = 6$, $c = 0$, $\pi = \frac{1}{2}$: $p_H = 5$, $p_L = 3$, $\pi_H = 12.5$, $\pi_L = 4.5$, $\delta^* = \frac{100}{150+18} = \frac{25}{42} \approx 0.595$. At $\pi = 1$ (no fluctuation) $\delta^* = \frac{1}{2}$.

(iii) Highest boom price for $\delta \in [\frac{1}{2}, \delta^*)$. Suppose the firms charge $p \le p_H$ in the high state, with static profit per firm $\varphi(p) = \frac{(p-c)(a_H - p)}{2}$, and p_L in the low state. The deviator undercuts at $p - \varepsilon$ and earns $2\varphi(p)$. The expected future profit is $\bar{\pi}(p) = \pi\varphi(p) + (1-\pi)\pi_L$. The high-state constraint is

$$\frac{\delta}{1-\delta}\big[\pi\varphi(p) + (1-\pi)\pi_L\big] \ge \varphi(p) \iff \varphi(p) \le \Phi(\delta) \equiv \frac{\delta(1-\pi)\pi_L}{1 - \delta(1+\pi)}.$$

($1 - \delta(1+\pi) > 0$ for $\delta < \delta^* < \frac{1}{1+\pi}$.) The low-state constraint $\delta\bar{\pi}(p) \ge (1-\delta)\pi_L$ holds because $\Phi(\delta) \ge \pi_L$ for $\delta \ge \frac{1}{2}$. Since φ rises up to p_H, the highest sustainable price is the lower root of $\varphi(p) = \Phi(\delta)$:

$$p(\delta) = \frac{(a_H + c) - \sqrt{(a_H - c)^2 - 8\Phi(\delta)}}{2},$$

which increases with δ, with $\Phi(\delta^*) = \pi_H$ giving $p(\delta^*) = p_H$. In the example $p(\frac{1}{2}) = 1$ (profit equals $\pi_L = 4.5$, so $p(\frac{1}{2}) < p_L$: at $\delta = \frac{1}{2}$ the high state can earn no more than the low state does), and $p(\delta)$ rises to 5 at $\delta = \frac{25}{42}$.

(d) Interpretation. Booms tempt firms more (a larger gain from cutting, while the future is unchanged), so for intermediate δ the boom price must be below monopoly: prices are countercyclical relative to the monopoly benchmark (Rotemberg–Saloner).

(e) Verification. `check_T3.py` (item 23) computes the prices, π_H, π_L, $\delta^* = \frac{25}{42}$ and the formula, confirms the low-state cutoff is smaller, that the high-state condition switches at δ^*, and, for $\delta \in \{\frac{1}{2}, \frac{11}{20}, \frac{57}{100}\}$, that the largest price on a grid of step $\frac{1}{2000}$ satisfying both constraints equals the formula for $p(\delta)$ (to 10^{-3}), with $p(\frac{1}{2}) = 1$. Gibbons's unofficial manual is not copied.

(f) Common mistakes.

- Using the state-contingent profit as the future value. After the next draw the future value is the *expected* $\bar{\pi}$, whichever state today is.
- Forgetting the low-state constraint when lowering the high-state price; it holds for $\delta \ge \frac{1}{2}$ here.

Answer. (i) $p_i = \frac{a_i + c}{2}$. (ii) $\delta^* = \dfrac{(a_H - c)^2}{(a_H - c)^2(1+\pi) + (1-\pi)(a_L - c)^2}$. (iii) For $\delta \in [\frac{1}{2}, \delta^*)$: $p(\delta) = \frac{1}{2}\big[(a_H + c) - \sqrt{(a_H - c)^2 - 8\Phi(\delta)}\big]$ with $\Phi(\delta) = \frac{\delta(1-\pi)(a_L - c)^2/8}{1 - \delta(1+\pi)}$, rising from the price that earns the low-state profit at $\delta = \frac{1}{2}$ to p_H at δ^*.

Item 24: Tadelis Exercise 11.7

Advanced · difficulty 4/5

Statement (verbatim from the list, item 24)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 11.7 "Legislative Bargaining Revisited", p. 237 [PDF 256]. Reproduced verbatim. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Legislative Bargaining Revisited:** Consider a finite T-period version of the Baron and Ferejohn legislative bargaining game with an odd number N of players and with a closed rule as described in Section 11.4.1.
>
> a. Find the unique subgame-perfect equilibrium for $T = 1$. Also find a Nash equilibrium that is not subgame perfect.
>
> b. Find the unique subgame-perfect equilibrium for $T = 2$ with a discount factor $0 < \delta \leq 1$. Also find a Nash equilibrium that is not subgame perfect.
>
> c. Compare what the first period's proposer receives in the subgame-perfect equilibrium you found in (b) to what a first-period proposer receives in the two-period, two-person Rubinstein-Ståhl bargaining game. What intuitively accounts for the difference?
>
> d. Compare the subgame-perfect equilibrium you found in (b) to the solution of the infinite-horizon model in Section 11.4.1. What intuitively accounts for the similarity?

Referred to in the statement. The Baron and Ferejohn closed-rule game and its infinite-horizon solution (Tadelis §11.4 and §11.4.1, pp. 229–231 [PDF p. 248–250]) and the two-period, two-person game of Section 11.2 (pp. 224–225 [PDF p. 243–244]) (reproduced from the book):

> In an important contribution, Baron and Ferejohn (1989) build on the Rubinstein-Ståhl model of bargaining and offer a framework for analyzing multilateral bargaining with a simple majority rule. The two main departures from the two-player model are as follows: First, there is an odd number N of players, where $\frac{N+1}{2}$ votes are needed to secure an agreement. Second, instead of players alternating their roles in a prespecified order, the bargaining protocol follows a random-assignment rule: in every period each player has an equal probability of being the proposer.
>
> Baron and Ferejohn consider two bargaining rules that to a large extent mimic common practices: the closed rule, under which proposals cannot be amended, and the open rule, under which they can. [. . .]
>
> **11.4.1 Closed-Rule Bargaining.** The closed-rule setting is very much an N-player version of the Rubinstein infinite-horizon model. When called upon to propose, the proposer makes an offer to the legislative body of the form $x = (x_1, x_2, \ldots, x_N)$, where x_i is the share of surplus offered to player i, with $\sum_{i=1}^{n} x_i \leq 1$. If a majority accepts the proposal then the game ends, and if not then the current "pie" is discounted according to the discount factor δ.
>
> It turns out that when there are three or more players in an N-player version of the Rubinstein infinite-horizon bargaining game then there are many possible subgame-perfect equilibria. What Baron and Ferejohn suggest is a simple way around this indeterminacy. They focus attention on *symmetric stationary equilibria* in which first, each player proposes the same split of the pie every time he is called to propose, regardless of the history of the game, and second, the respondents vote based only on the current proposal as it compares to their expectations about future proposals. [. . .]
>
> Let v be the expected payoff for any player i to play the subgame-perfect equilibrium of this game at the beginning of any stage, and consider a player i who is choosing to respond to a proposal that awards him x_i. If he rejects the proposal then he expects to get v in the next period, and hence he will accept the proposal if and only if $x_i \geq \delta v$. As a consequence, any player who plays the role of a proposer anticipates that he must propose at least δv to at least $\frac{N+1}{2}$ other players.

Given this observation, the proposer's best response is to choose exactly $\frac{N-1}{2}$ other players and offer them exactly δv each, keeping the rest of the pie for himself. [...] Hence the portion of the pie that the proposer keeps for himself in the symmetric subgame-perfect equilibrium is

$$k = 1 - \frac{N-1}{2}\delta v. \tag{11.6}$$

To maintain symmetry, each proposer chooses his $\frac{N-1}{2}$ coalition partners randomly from the remaining $N-1$ players, implying that each of the $N-1$ responders gets chosen to be part of the coalition with probability $\frac{1}{2}$.

The equilibrium analysis leads to the computation of v. At the beginning of the game (and of every stage after rejection), each player expects to play in one of three roles. He can either be the proposer and receive k, which occurs with probability $\frac{1}{N}$, or be a responder, which occurs with probability $\frac{N-1}{N}$. If he is a responder, then with probability $\frac{1}{2}$ he will be in the winning coalition and receive δv, or he can be outside the coalition and receive zero. This implies that

$$v = \frac{k}{N} + \frac{N-1}{2N}\delta v. \tag{11.7}$$

Combining (11.6) and (11.7) we obtain that $v = \frac{1}{N}$. [...]

It is more interesting to focus attention on the value of k, the share obtained by the proposer, which is

$$k(N) = 1 - \delta\left(\frac{N-1}{2N}\right). \tag{11.8}$$

The two-period, two-person game (Section 11.2). As Figure 11.2 demonstrates, the second round is a one-round bargaining game, which implies that if it is reached then player 2 gets the whole pie in any subgame-perfect equilibrium. Continuing with backward induction, and assuming that $\delta \leq 1$, we can find the subgame-perfect equilibrium for the two-round game as follows. In the first round player 2 knows that in the next (and last) round he will receive the whole pie, which will be worth only δ because of discounting. Thus if he is offered less than δ for himself (player 1 offers $x > 1-\delta$) then player 2 should reject the offer and get the whole pie in the next period. Hence the unique sequentially rational strategy for player 2 at the accept/reject stage of the first round is to accept any offer of $x \leq 1-\delta$. Moving up the tree to the node at which player 1 makes his first offer, he realizes that any offer $x > 1-\delta$ will be rejected, while any offer $x \leq 1-\delta$ will be accepted. Thus player 1's best response is to offer $x = 1-\delta$, leading to a division of surplus equal to $(v_1, v_2) = (1-\delta, \delta)$.

(a) What is asked. In a T-period Baron–Ferejohn game with an odd number N of players and a closed rule: (a) the SPE for $T = 1$ and a Nash equilibrium that is not subgame perfect; (b) the same for $T = 2$ with discount factor $\delta \in (0, 1]$; (c) compare the first proposer's share with two-person Rubinstein–Ståhl bargaining; (d) compare with the infinite-horizon solution of Section 11.4.1 and explain the similarity.

(b) Method. Tadelis §11.4.1 (pp. 229–232). Each period a proposer is drawn uniformly, proposes shares $x = (x_1, \ldots, x_N)$ with $\sum x_i \leq 1$, and a majority of $\frac{N+1}{2}$ votes (the proposer's included) passes it. Otherwise the game moves to the next period with the pie discounted by δ. If the last period fails, all get 0. A voter accepts iff his share is at least the discounted value δv of waiting, where v is his expected payoff at the start of a period.

(c) Solution. **(a)** $T = 1$**.** A voter who is offered $x_i \geq 0$ accepts (rejecting gives 0). The proposer needs $\frac{N-1}{2}$ other votes and buys them with 0 each, keeping the whole pie: proposer 1, everybody else 0. Each player's expected payoff is $v_1 = \frac{1}{N}$ (he is the proposer with probability $\frac{1}{N}$).

A non-subgame-perfect Nash equilibrium. Every proposer offers the equal split $\frac{1}{N}$ to all, and every voter votes yes iff the proposal is *exactly* the equal split. On the path the equal split passes. A proposer who deviates to any other division gets at most his own vote ($1 < \frac{N+1}{2}$ for $N \geq 3$), so it fails and pays $0 < \frac{1}{N}$. No voter's vote changes the outcome, so no one deviates. It is Nash, with expected payoff $\frac{1}{N}$ to everyone. It is not subgame perfect: a voter offered a positive amount (or any $x \neq \frac{1}{N}$) who votes no gets $0 < x$ in the last period, so rejecting is not a best reply there.

(b) $T = 2$**.** After a failure in period 1 each player expects $v_1 = \frac{1}{N}$ from period 2, so a voter accepts iff $x_i \geq \frac{\delta}{N}$. The proposer chooses $\frac{N-1}{2}$ others at random and offers $\frac{\delta}{N}$ to each, keeping

$$k_2 = 1 - \frac{N-1}{2} \cdot \frac{\delta}{N} = 1 - \frac{\delta(N-1)}{2N}.$$

It passes in period 1. Expected payoff of a player: $\frac{1}{N}k_2 + \frac{N-1}{N} \cdot \frac{1}{2} \cdot \frac{\delta}{N} = \frac{1}{N}$, the same $v_2 = \frac{1}{N}$.

A non-SPE Nash equilibrium. The equal-split rule of (a) in period 1 (voters accept only the equal split) and the SPE of $T = 1$ in period 2. A deviating proposer's offer is rejected and he gets $\delta \cdot \frac{1}{N} < \frac{1}{N}$; voters are not pivotal. It is not subgame perfect, for the same reason as before: a voter offered $x_i > \frac{\delta}{N}$ rejects it, though waiting is worth only $\frac{\delta}{N}$.

(c) Comparison with two-person bargaining. With $T = 2$ and two players the proposer gets $1 - \delta$. Here the proposer keeps $1 - \frac{\delta(N-1)}{2N} > 1 - \delta$ for $N \geq 3$, because $\frac{N-1}{2N} < \frac{1}{2} < 1$. In the two-person game the responder who rejects becomes the sole proposer and takes everything in period 2, which makes him expensive to satisfy. In the legislature a voter who rejects is only the proposer with probability $\frac{1}{N}$, so his continuation value is $\frac{\delta}{N}$ and each needed vote is cheap, and the proposer needs only a bare majority.

(d) Comparison with the infinite horizon. The symmetric stationary equilibrium has $v = \frac{1}{N}$ and the proposer keeps $k = 1 - \frac{N-1}{2}\delta v = 1 - \frac{\delta(N-1)}{2N}$, identical to k_2. By symmetry and random recognition, every player's expected payoff at the start of *any* period is $\frac{1}{N}$, whatever the horizon, so the price of a vote ($\delta v = \frac{\delta}{N}$) does not depend on how many periods remain. A longer horizon changes nothing that matters to the voters' reservation values, which is why $T = 2$ already reproduces the infinite-horizon split.

(d) Interpretation. Random recognition spreads the proposer power evenly, making the continuation value independent of the horizon. In the two-person game, who proposes last decides everything, which is why finite and infinite horizons differ.

(e) Verification. `check_T3.py` (item 24) iterates $v_T = \frac{k}{N} + \frac{N-1}{N} \cdot \frac{1}{2}\delta v_{T-1}$ for $N \in \{3, 5, 7, 9\}$, $\delta \in \{\frac{1}{2}, \frac{9}{10}, 1\}$ and $T \leq 7$ and finds $v_T = \frac{1}{N}$ each time, that the shares add to 1, that the infinite-horizon fixed point is $\frac{1}{N}$ with $k_\infty = k_2$, and that $k_2 > 1 - \delta$. Tadelis prints no answer.

(f) Common mistakes.

- Offering votes the whole current share or an equal share: the price of a vote is the discounted continuation value $\frac{\delta}{N}$, not $\frac{1}{N}$.
- Buying more than $\frac{N-1}{2}$ votes. A minimal winning coalition is the cheapest.

Answer. (a) $T = 1$: proposer takes 1, others 0; each player expects $\frac{1}{N}$; a non-SPE Nash equilibrium is "everybody offers and votes for the equal split only". (b) $T = 2$: the proposer offers $\frac{\delta}{N}$ to $\frac{N-1}{2}$ others and keeps $1 - \frac{\delta(N-1)}{2N}$; each player expects $\frac{1}{N}$. (c) That exceeds $1 - \delta$ in the two-person game, as the votes are cheaper. (d) It equals the infinite-horizon proposer share, because every player's continuation value is $\frac{1}{N}$ at any horizon.

Item 25: Cabral Exercise 9.19

Advanced · difficulty 5/5

Statement (verbatim from the list, item 25)

Source. Cabral (2017), *Introduction to Industrial Organization*, 2nd ed., Ch. 9 Challenging exercises, Exercise 9.19 "Secret price cuts", p. 245 [PDF 257]. Reproduced verbatim, with the margin note (s) it cites (the book's "price cups" kept). Answer in the source: none in the book.

> **9.19. Secret price cuts.** This exercise formalizes the model of secret price cups presented in Section 9.2. Suppose that all consumers are willing to pay u for the (homogeneous) product sold by two duopolists. In each period, demand can be high (probability $1-\alpha$) or low (probability α). When demand is high, $h=1$ units can be sold at price u (or any lower price). When demand is low, only $l=0<h$ units can be sold. The probability that demand is high or low in each period is independent of what it was in the previous period. Moreover, *firms are unable to observe the state of market demand*; all they can observe is whether *their own* demand is high or low. Finally, for simplicity, assume that production costs are zero.
>
> Consider the following equilibrium strategies. Firms start by setting $p=u$. If they receive a positive demand (namely, $\frac{1}{2}$), then they continue to set $p=u$, that is, they remain in the "cooperative phase". If however one of the firms (or both) receives zero demand, then both firms enter into a "price war:" they set $p=0$ during T periods and, after this period, revert to $p=u$ again (the cooperative phase).[s] Let V be the net present value of a firm in equilibrium (starting in a period where collusion takes place).
>
> (a) Show that:
>
> $$V=(1-\alpha)\left(\frac{u}{2}+\delta V\right)+\alpha\,\delta^{T+1}V$$
>
> (b) Show that the best a firm can do by deviating is to get:
>
> $$V'=(1-\alpha)\,u+\delta^{T+1}V$$
>
> (c) Show that the condition that the prescribed strategy constitutes an equilibrium is given by:
>
> $$1\leq 2\,(1-\alpha)\,\delta+(2\,\alpha-1)\,\delta^{T+1}$$
>
> (d) What is then the optimal equilibrium?
>
> *s.* Notice that, if a firm receives zero demand, then it is common knowledge that a price war is going to start, that is, it is common knowledge that one of the firms receives zero demand. In fact, either demand is low, in which case both firms receive zero demand, or one of the firms deviates from $p=u$, in which case the deviating firm knows that the rival receives zero demand.

(a) What is asked. Two duopolists face demand $h=1$ (probability $1-\alpha$) or $l=0$ (probability α) at reservation price u, with zero cost, and each firm sees only its own demand. They set $p=u$ until a firm gets zero demand, then $p=0$ for T periods, then $p=u$ again. Show (a) the value V; (b) the deviation value V'; (c) the equilibrium condition; (d) the optimal equilibrium.

(b) Method. Cabral §9.2 (pp. 222–227; secret price cuts, Green–Porter). The strategy is an equilibrium iff the value of cooperation V is at least the value V' of the best deviation (one-stage deviation principle). The price war itself is the static equilibrium ($p=0$), so it needs no check.

(c) Solution. **(a) V.** In a cooperative period, with probability $1-\alpha$ demand is high and each firm sells $\frac{1}{2}$ at u, earning $\frac{u}{2}$ and staying in the cooperative phase (value δV next period). With probability α demand is low: profit 0, and the price war follows for T periods of zero profit, after which cooperation returns, a value $\delta^{T+1}V$ viewed from today. Hence

$$V = (1-\alpha)\Big(\frac{u}{2} + \delta V\Big) + \alpha\,\delta^{T+1}V \quad\Longrightarrow\quad V = \frac{(1-\alpha)u/2}{1-(1-\alpha)\delta - \alpha\delta^{T+1}}.$$

(b) V'. A deviator undercuts u slightly. If demand is high (probability $1-\alpha$) he sells 1 unit at about u, earning u, and the rival, who then gets zero demand, starts the price war. If demand is low (probability α) he earns 0 either way and the war starts anyway. In both cases the future is $\delta^{T+1}V$ (war and return):

$$V' = (1-\alpha)u + \delta^{T+1}V.$$

(c) Condition. $V \geq V'$ means $V\,(1-\delta^{T+1}) \geq (1-\alpha)u$. From (a), $(1-\alpha)u = 2V\,[\,1-(1-\alpha)\delta - \alpha\delta^{T+1}\,]$. Substituting and dividing by $V > 0$:

$$1-\delta^{T+1} \geq 2 - 2(1-\alpha)\delta - 2\alpha\delta^{T+1} \iff 1 \leq 2(1-\alpha)\delta + (2\alpha-1)\delta^{T+1}.$$

(d) Optimal equilibrium. The planner chooses T to maximise V, which is increasing in δ^{T+1}, so a *shorter* war is better, subject to (c). Write $g(T) = 2(1-\alpha)\delta + (2\alpha-1)\delta^{T+1}$.

- If $\alpha \geq \frac{1}{2}$: $g(T) \leq \delta < 1$ for every $T \geq 0$: no such equilibrium exists. (Wars are triggered too often.)
- If $\alpha < \frac{1}{2}$: g increases in T and tends to $2(1-\alpha)\delta$, so an equilibrium exists iff $\delta > \frac{1}{2(1-\alpha)}$, and the optimal one uses the *smallest* T with $\delta^{T+1} \leq \dfrac{2(1-\alpha)\delta - 1}{1-2\alpha}$.

Example: $\alpha = \frac{1}{5}$, $\delta = \frac{9}{10}$: $g(1) = 0.954 < 1 \leq g(2) = 1.0026$, so $T^* = 2$ and $V = 2.98$ (with $u = 1$). Equilibrium price wars occur on the path (after bad draws) even though nobody cheats: they are the cost of imperfect monitoring.

(d) Interpretation. A price war must be long enough to deter secret cuts and short enough not to waste surplus. The optimal war is the shortest one that deters, and it happens with positive probability on the equilibrium path.

(e) Verification. `check_T3.py` (item 25) solves (a) symbolically, derives $(V - V')$ times the denominator and shows it equals $g(T) - 1$ times $\frac{(1-\alpha)u}{2}$ for $T = 1, 2, 3$, confirms $V = 2.98$ by value iteration over the Markov chain of phases ($T = 2$, $\alpha = 0.2$, $\delta = 0.9$), shows that (c) is never met for $\alpha \geq \frac{1}{2}$, and finds $T^* = 2$ by search together with $\lceil \ln r / \ln \delta \rceil - 1$ for $r = \frac{2(1-\alpha)\delta - 1}{1-2\alpha}$. Cabral prints no answer.

(f) Common mistakes.

- Treating the war as a punishment for cheating only: it also follows low demand, which is why the equilibrium has wars on the path.
- Maximising T: a longer war lowers V (more periods of zero profit).

Answer. (a) $V = \frac{(1-\alpha)u/2}{1-(1-\alpha)\delta-\alpha\delta^{T+1}}$. (b) $V' = (1-\alpha)u + \delta^{T+1}V$. (c) $V \geq V'$ is equivalent to $1 \leq 2(1-\alpha)\delta + (2\alpha-1)\delta^{T+1}$. (d) Exists iff $\alpha < \frac{1}{2}$ and $\delta > \frac{1}{2(1-\alpha)}$; the optimal war has the smallest T satisfying (c), e.g. $T^* = 2$ for $\alpha = \frac{1}{5}$, $\delta = 0.9$.

Item 26: Tadelis Exercise 10.10

Advanced · difficulty 4/5

Statement (verbatim from the list, item 26)

Source. Tadelis (2013), *Game Theory: An Introduction*, Exercise 10.10 "Law Merchants Revisited", p. 219 [PDF 238]. Reproduced verbatim. Figure 10.2 (p. 206) is not part of the exercise text; it is added because Section 10.5.2, which the statement cites, builds the game on it. Answer in the source: none in the book (the preface, p. xiii, says about half are solved on Princeton UP's website; not checked).

> **Law Merchants Revisited:** Consider the three-person game described in Section 10.5.2. A subgame-perfect equilibrium was constructed with a bond equal to 2, a fee equal to $w = 0.1$ was paid by every player P_2^t to player 3, and it was shown that it is indeed an equilibrium for any discount factor $\delta \geq 0.95$. Show that a similar equilibrium exists in which players P_1^t trust players P_2^t who post bonds, players P_2^t post bonds and cooperate, and player 3 follows the contract in every period, for any discount factor $0 < \delta < 1$.

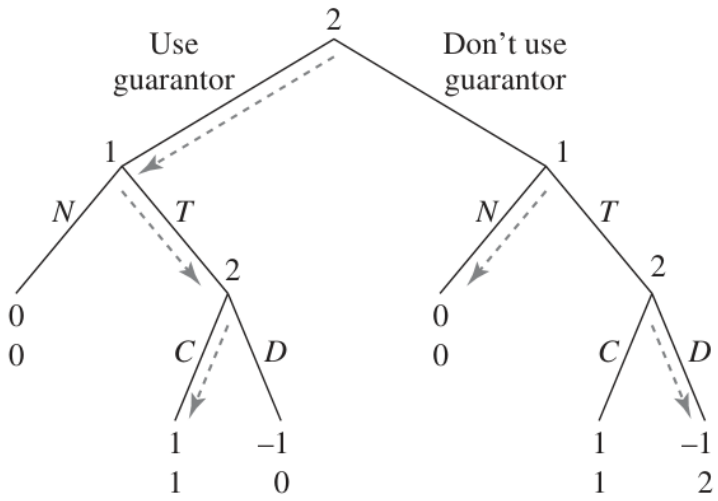


FIGURE 10.2 Trust game with a bond option.

Figure 10.2 Trust game with a bond option (Tadelis p. 206; added for reference).

(a) What is asked. In the three-person game of Section 10.5.2 (a bond $B = 2$ and a fee $w = 0.1$ sustain cooperation for $\delta \geq 0.95$), show that a similar equilibrium (trust, bond, cooperation, guarantor honest in every period) exists for any discount factor $0 < \delta < 1$.

(b) Method. The structure of §10.5.2 (pp. 205–207). The trust game of Figure 10.1: no trust $(0, 0)$; trust and cooperate $(1, 1)$; trust and defect $(-1, 2)$ for players (P_1^t, P_2^t). With a bond B posted with player 3, a defector loses B (Figure 10.2 uses $B = 2$). The guarantor (player 3) lives for ever and discounts at δ. Footnote 10 (p. 207) says the bond need only exceed 1 and that P_2^t is willing to pay any fee up to 1. We choose B and w and verify three incentive constraints, using the book's conventions: P_2^t pays the fee in every period in which the guarantor follows the contract (including after his own defection, when the guarantor keeps the bond), and the guarantor's alternative is to keep the bond after a cooperation, which earns B in that period and ends all future business.

(c) Solution. Strategies (as in the text): P_1^t trusts iff a bond was posted. P_2^t posts a bond in period t if the guarantor has followed the contract in all earlier periods, and then cooperates; otherwise he does not post and defects. The guarantor returns the bond iff P_2^t cooperated.

(1) P_2^t cooperates, given the bond. With the bond she gets $1 - w$ from C and $2 - B - w$ from D (she pays the fee in both cases, the bond is lost after D). Hence C is optimal iff

$$B \geq 1,$$

independently of w.

(2) P_2^t uses the guarantor. Without a bond she would defect and P_1^t would not trust, so she gets 0. With it she gets $1 - w$. She joins iff $w \leq 1$ (strictly, $w < 1$).

(3) The guarantor stays honest. Following the contract earns w every period; keeping the bond after a cooperation earns B once and then no one posts a bond again. He complies iff

$$\frac{w}{1-\delta} \geq B \iff w \geq (1-\delta)B.$$

(The book's check is the same with $B = 2$, $w = 0.1$: $\frac{0.1}{1-\delta} \geq 2$, i.e. $\delta \geq 0.95$.)

Choice for any δ. The three conditions need $B \geq 1$ (to avoid indifference take $B > 1$), $(1-\delta)B \leq w < 1$. For B slightly above 1 the interval $\big[(1-\delta)B,\, 1\big)$ is non-empty for *every* $\delta > 0$, since $(1-\delta)B < 1$ when $B < \frac{1}{1-\delta}$. A concrete choice:

$$B = 1 + \frac{\delta}{2}, \qquad w = 1 - \frac{\delta}{2}.$$

Then $B > 1$ (cooperation is strictly optimal), $w < 1$ (she strictly prefers to use the guarantor, with surplus $\frac{\delta}{2}$) and $\frac{w}{1-\delta} = \frac{1-\delta/2}{1-\delta} > 1 + \frac{\delta}{2} = B$ (the guarantor strictly prefers to comply). Off the path nothing changes: after the guarantor cheats nobody posts a bond and (N), D are stage equilibrium choices.

(d) Interpretation. The fee need only compensate the guarantor for the one-period gain from keeping a bond that is just large enough to deter defection (B close to 1, the defector's gain). Making the bond small and the fee large shrinks the guarantor's temptation relative to the stream of fees, so even very impatient guarantors are honest. The limit is the fee ceiling $w < 1$ (what the clients are willing to pay) and $B \geq 1$.

(e) Verification. `check_T3.py` (item 26) checks $\frac{0.1}{1-\delta} = 2$ at $\delta = 0.95$ and verifies the three inequalities for $B = 1 + \frac{\delta}{2}$, $w = 1 - \frac{\delta}{2}$ at every $\delta = \frac{k}{100}$, $k = 1, \ldots, 99$. Tadelis prints no answer.

(f) Common mistakes.

- Fixing the bond at 2 and the fee at 0.1: with those numbers $\delta \geq 0.95$ is needed.
- Charging a fee above 1: P_2^t then prefers no trade.

Answer. Take bond $B = 1 + \frac{\delta}{2}$ and fee $w = 1 - \frac{\delta}{2}$. Then P_2^t cooperates ($B \geq 1$), joins the guarantor ($w < 1$) and the guarantor complies ($\frac{w}{1-\delta} \geq B$), for every $\delta \in (0, 1)$. In general the conditions are $B \geq 1$, $w \leq 1$ and $w \geq (1-\delta)B$.

Item 27: Gibbons Problem 2.17

Advanced · difficulty 4/5

Statement (verbatim from the list, item 27)

Source. Gibbons (1992), *Game Theory for Applied Economists*, Ch. 2 Problems, Section 2.3, Problem 2.17, pp. 136–137 [PDF 143–144]. Reproduced verbatim. Answer in the source: none (Gibbons prints no answers).

> 2.17. Consider the following infinite-horizon game between a single firm and a sequence of workers, each of whom lives for one period. In each period the worker chooses either to expend effort and so produce output y at effort cost c or to expend no effort, produce no output, and incur no cost. If output is produced, the firm owns it but can share it with the worker by paying a wage, as described next. Assume that at the beginning of the period the worker has an alternative opportunity worth zero (net of effort cost) and that the worker cannot be forced to accept a wage less than zero. Assume also that $y > c$ so that expending effort is efficient.
>
> Within each period, the timing of events is as follows: first the worker chooses an effort level, then output is observed by both the firm and the worker, and finally the firm chooses a wage to pay the worker. Assume that no wage contracts can be enforced: the firm's choice of a wage is completely unconstrained. In a one-period game, therefore, subgame-perfection implies that the firm will offer a wage of zero independent of the worker's output, so the worker will not expend any effort.
>
> Now consider the infinite-horizon problem. Recall that each worker lives for only one period. Assume, however, that at the beginning of period t, the history of the game through period $t-1$ is observed by the worker who will work in period t. (Think of this knowledge as being passed down through the generations of workers.) Suppose the firm discounts the future according to the discount factor δ per period. Describe strategies for the firm and each worker in a subgame-perfect equilibrium in the infinite-horizon game in which in equilibrium each worker expends effort and so produces output y, provided the discount factor is high enough. Give a necessary and sufficient condition for your equilibrium to exist.

(a) What is asked. An infinitely lived firm faces a sequence of one-period workers. A worker chooses effort (output y, cost $c < y$) or not; the firm then chooses a wage, with no enforceable contracts; each worker observes the whole past. Describe strategies for the firm and the workers forming an SPE in which every worker works, and give a necessary and sufficient condition.

(b) Method. A relational contract through trigger strategies (Gibbons §2.3.D, pp. 107–112, efficiency wages; the firm is the long-lived player). The worker's outside option is 0 and a wage cannot be negative, so a worker works only if the wage he expects, net of the effort cost, is at least 0. The worst the workers can do to the firm is never work, which gives it 0.

(c) Solution. **Strategies.** Fix a wage $w^* \in [c, y)$.

- *Worker t:* work iff in every earlier period in which output was produced the firm paid w^* (and, in period 1, always); otherwise shirk.
- *Firm:* after output pay w^* as long as it has always done so; after any failure to pay w^* following output, pay 0 for ever.

Worker. If he works he expects w^* and gets $w^* - c \geq 0$, the value of his outside option, so working is optimal. After the firm has once cheated, the worker expects wage 0 and shirks (working would cost $c > 0$ for nothing). That is a best reply.

Firm. On the path it gets $y - w^*$ per period: $\frac{y-w^*}{1-\delta}$. If it pays 0 after output it keeps y once and then no worker ever works again (0). It pays iff

$$\frac{y - w^*}{1 - \delta} \geq y \iff w^* \leq \delta y.$$

After a deviation the firm pays 0 and no output is produced, which is a stage best reply (a wage after zero output costs it), so the punishment phase is itself an equilibrium.

Condition. A wage w^* must satisfy $c \leq w^* \leq \delta y$, so the equilibrium exists iff

$$\delta \geq \frac{c}{y}.$$

Necessity. In any SPE in which workers work, the equilibrium wage must be at least c (workers can refuse). The firm can always deviate to 0, and the harshest punishment it can face is zero profit for ever, because workers can always shirk. So the equilibrium wage is at most δy, which is impossible if $\delta < \frac{c}{y}$. Hence $\delta \geq \frac{c}{y}$ is also necessary.

(d) Interpretation. The firm honours an unenforceable wage promise because reneging ends the relationship. The wage that induces effort (c) is the minimum a worker accepts, and the bound δy is the most the firm will pay without being tempted to renege. The relationship is possible iff effort is cheap relative to the firm's patience ($\frac{c}{y} \leq \delta$): output must exceed the effort cost with enough margin, discounted.

(e) Verification. `check_T3.py` (item 27) checks, for $(y, c) \in \{(10, 4), (10, 9), (5, 1)\}$ and all $\delta = \frac{k}{100}$, that a wage in $[c, \delta y]$ exists iff $\delta \geq c/y$, that every such wage satisfies the firm's constraint, and that when $\delta < c/y$ no wage $\geq c$ does. Gibbons's unofficial manual is not copied.

(f) Common mistakes.

- Specifying the firm's strategy only on the path. SPE requires what happens after a failure to pay.
- Setting the wage at y or at $y - c$: the cap is δy, and a wage above y would not be paid.

Answer. Workers work iff the firm has always paid w^* after output; the firm pays w^* as long as it always has, and 0 for ever otherwise, with $c \leq w^* \leq \delta y$. The firm complies iff $\frac{y-w^*}{1-\delta} \geq y$, and the worker iff $w^* \geq c$. Such w^* exists, and the SPE exists, iff $\delta \geq \dfrac{c}{y}$ (also necessary).

Item 28: Milgrom & Roberts Ch. 8, Mathematical Exercise 1 Advanced · difficulty 4/5

Statement (verbatim from the list, item 28)

Source. Milgrom & Roberts (1992), *Economics, Organization and Management*, Ch. 8 Exercises, Mathematical Exercise 1, p. 285 [PDF 300]. Reproduced verbatim (dollar amounts written with an escaped sign, `\$`). Answer in the source: none in the book.

1. In the context of the efficiency wage model, suppose that N is given by $(1-\delta^t)/(1-\delta)$, where δ is a discount factor giving the value today of a dollar to be received a year from now and t is the number of years the worker anticipates remaining with the firm. Thus, N is a measure of the horizon over which the employee is concerned, appropriately discounted. Suppose the worker expects to be employed for 5 years, that is, $t = 5$, and evaluates future income using $\delta = 0.9$, so that N is approximately equal to 4.1. The worker's opportunity wage ($\overline{w}$) is \$20,000 per year, and he or she can make an immediate gain of \$3,000 by leaking the firm's trade secrets to a competitor. The firm can monitor the worker to prevent this, but doing so is costly: raising the probability p of catching misbehavior by an amount Δp costs \$1,500 Δp.

   Formulate the firm's problem in selecting the monitoring intensity p and the wage w and solve for the optimal values.

   Suppose now that the threat of a takeover or bankruptcy endangers the employee's future with the firm, so that the employee now expects to be employed only one more year. Will the contract you found above still deter cheating? Why or why not?

   More generally, show that in the efficiency wage model, a lengthened horizon allows providing incentives with a lower wage. That is, show that the optimal wage level $w^*(N)$ is a nonincreasing function of N provided the marginal cost of monitoring is positive and increasing in p, that is, provided $M'(p) > 0$ and $M''(p) > 0$ for all p.

Referred to in the statement. The efficiency wage model of Chapter 8: "The Shapiro-Stiglitz Model" and "A Mathematical Example: Comparative Statics for Efficiency Wages" (Milgrom & Roberts pp. 250–251 and 254 [PDF p. 265–266 and 269]) (reproduced from the book):

> Let w be the wage the employee is paid in his or her current job (say, \$50,000), and let $\overline{w}$ be the wage the employee could get if he or she looked for another job in the market after being fired from the current job (say, \$40,000). The wage $\overline{w}$ is assumed to be discounted to take into account the cost of searching for a new job, including any period of unemployment during the search, the cost of submitting applications, and so on. Let g be the amount that the employee could gain by cheating on the job, regardless of whether he or she is detected. [...] Let p be the probability that the cheating, whatever form it may take, is detected. Let N be the multiplier to express the long-term value of this relationship, adjusting for both interest and the number of periods during which the worker is employed by the firm if not caught cheating. If the employment relation lasts only one period and is never repeated, then $N = 1$. If the employee is hired from time to time and expects to be rehired in the future only if he or she has never cheated the employer, then N is larger than 1. If there is a continuing employment relation between the worker and the employer, then N may be much larger than 1.
>
> Putting ethics and morality aside momentarily, it will serve the employee's narrow personal interest to cheat if
>
> $$g > p(w - \overline{w})N \tag{8.1}$$
>
> that is, if the gain from cheating exceeds the probability that the cheating is detected times loss of income from being fired. [...]
>
> Let $M(p)$ be the monitoring cost incurred in every accounting period when the probability of detection is p. If the company wants to deter cheating, it can minimize its total cost in each period by choosing its wage and monitoring policy to solve:

$$\begin{aligned} &\underset{p,\,w}{\text{Minimize}}\; M(p) + w \\ &\text{subject to } p(w - \overline{w})N \geq g \end{aligned} \tag{8.2}$$

Further practice (not in the tiers). Source lines only; the statements are in the previous

Lecture 4 list (`git show fe47ead:Listas/pratica-lecture-04.md`), where they were verified.

- Muñoz-Garcia Ch. 4 Ex. 16 (pp. 198–199, "Two-Period Bilateral Bargaining Game", with Fig. 4.35) and Ex. 17 (pp. 199–200, "Alternating Bargaining with a Twist"): costly offers and a shrinking surplus, no discounting.

- Tadelis 11.4 (p. 236, "Constant Delay Cost") and 11.6 (p. 237, "Asymmetric Patience 2"): delay costs instead of discounting; Rubinstein with $\delta_1 \neq \delta_2$.

- MWG 9.B.12 (p. 303) and 9.B.13 with 9.B.7 (pp. 302–303): the infinite-horizon game with $\delta_1 \neq \delta_2$; a cost per offer instead of discounting.

- Muñoz-Garcia Ch. 6 Ex. 8 (p. 303, "Infinite-Horizon Bargaining Game Between Three Players").

Optional extras (collusion, reputation, wages).

- Tadelis 10.1 (p. 215, discount cards and tacit collusion, discursive), 10.7 (pp. 217–218, the bartender's quality reputation, a twin of item 18) and 10.11 (p. 219, the brand-name equilibrium of §10.5.3).

- Cabral 9.2 (p. 242, the largest Bertrand cartel at $\delta = .8$), 9.5 (p. 242, are price wars irrational?), 9.18 (p. 245, multimarket contact again) and 9.20 (p. 246, observed demand fluctuations: Cabral's version of item 23).

- Cabral 9.17 (p. 245): the book asks to show that the minimum discount factor $\bar{\delta}$ is *decreasing* in n; with $\bar{\delta} = 1 - 1/n$ (Cabral p. 227) it is increasing. Work it as "show how $\bar{\delta}$ moves with n".

- Milgrom & Roberts Mathematical Exercise 2 (pp. 285–286, voluntary quits in the efficiency-wage model) and Food for Thought 1 (p. 284, wage surveys).

- Gibbons 2.3 (p. 131, Rubinstein with $\delta_1 \neq \delta_2$) and 2.16 (p. 136, a firm and a union in an infinitely repeated wage-employment game).

(a) What is asked. In the efficiency-wage model with $N = (1-\delta^t)/(1-\delta)$, $t = 5$, $\delta = 0.9$ ($N \approx 4.1$), opportunity wage $\bar{w} = 20\,000$, gain from leaking secrets $g = 3\,000$ and monitoring cost $1\,500$ per unit of detection probability: (1) formulate and solve for the optimal p and w; (2) say whether the contract still deters cheating if the horizon falls to one year; (3) show that $w^*(N)$ is nonincreasing in N when $M' > 0$ and $M'' > 0$.

(b) Method. The minimum-cost implementation problem (pp. 254–255, formulas 8.2–8.6):

$$\min_{p,w}\; M(p) + w \quad \text{s.t.} \quad p\,(w - \bar{w})\,N \geq g, \qquad 0 \leq p \leq 1.$$

The constraint binds, so $w = \bar{w} + \frac{g}{Np}$ (8.3) and the problem is $\min_p M(p) + \bar{w} + \frac{g}{Np}$ with first-order condition $M'(p^*) = \frac{g}{Np^{*2}}$ (8.5).

(c) Solution. (1) Optimum. $N = \frac{1-0.9^5}{0.1} = 4.0951$. With $M(p) = 1\,500\,p$ ($M' = 1\,500$):

$$1\,500 = \frac{3\,000}{Np^2} \iff p^* = \sqrt{\frac{2}{N}} = 0.699, \qquad w^* = 20\,000 + \frac{3\,000}{4.0951 \times 0.699} \approx 21\,048.$$

The second-order condition $\frac{2g}{Np^3} > 0$ holds. With $N = 4.1$, $p^* = 0.698$. The monitoring cost $1\,500\,p^* \approx 1\,048$ equals the wage premium $w^* - \bar{w}$: at the optimum the firm spends the same on monitoring as on the premium.

(2) One more year. With $N = 1$ the old contract gives an expected loss $p^*(w^* - \bar{w}) \cdot 1 = 0.699 \times 1\,048 \approx 733 < 3\,000$: cheating pays, so the contract no longer deters. Re-optimising: $p^* = \sqrt{2} > 1$ is infeasible, so $p = 1$ (the corner) and $w = 20\,000 + 3\,000 = 23\,000$. A shorter horizon makes cheating tempting, and deterrence needs much higher pay (or monitoring).

(3) $w^*(N)$ nonincreasing. The first-order condition gives $p^{*2}M'(p^*) = \frac{g}{N}$. The left side $\phi(p) = p^2M'(p)$ is strictly increasing in p when $M' > 0$ and $M'' > 0$, so $p^*(N)$ is the unique solution and is *decreasing* in N (the right side falls as N rises); this is the text's Figure 8.1. At the optimum $\frac{g}{Np^*} = p^*M'(p^*)$ by the first-order condition, so

$$w^*(N) = \bar{w} + p^*(N)\, M'\big(p^*(N)\big).$$

The function $p \mapsto pM'(p)$ is increasing ($M' > 0$, $M'' > 0$) and $p^*(N)$ is decreasing in N, hence $w^*(N)$ is nonincreasing in N. If $p^* \geq 1$ at some N, the corner $p = 1$ gives $w = \bar{w} + \frac{g}{N}$, also nonincreasing, and the two branches join continuously.

(d) Interpretation. A longer horizon means more rent to lose and more trust, so incentives can be provided with a lower wage premium and less monitoring. This is why long-tenure employees are paid less per unit of trust than short-horizon ones, and why takeovers or bankruptcy threats, which shorten N, erode honesty.

(e) Verification. `check_T3.py` (item 28) obtains $N = 4.0951$, $p^* = \sqrt{2/N} = 0.69886$, $w^* = 21\,048.2$ (and $p^* = 0.69843$ at $N = 4.1$), checks the second-order condition, that monitoring cost equals the wage premium, that $\sqrt{2} > 1$ at $N = 1$ (so $p = 1$ and $w = 23\,000$), the old contract's expected loss of 732.6, and, for $M(p) = mp^2$ with $m \in \{1, 100, 1\,500\}$, that $p^*(N)$ and $w^*(N)$ are strictly decreasing on a grid, that $w^* = \bar{w} + p^*M'(p^*)$, and that $Np^*(N)$ increases. The book prints no answer.

(f) Common mistakes.

- Ignoring the cap $p \leq 1$ when $N = 1$: the unconstrained $p^* = \sqrt{2}$ is not a probability.
- Proving monotonicity of w^* from $w = \bar{w} + g/(Np)$ alone: p^* also depends on N. The sign is settled by $w^* = \bar{w} + p^*M'(p^*)$ and the fall of p^*.

Answer. (1) $p^* = \sqrt{2/N} \approx 0.70$ and $w^* \approx \$21\,048$. (2) With $N = 1$ the old contract does not deter (expected loss $\approx \$733 < \$3\,000$); the new optimum is $p = 1$ and $w = \$23\,000$. (3) $M'(p^*) = g/(Np^{*2})$ implies $p^*(N)$ decreasing and $w^*(N) = \bar{w} + p^*M'(p^*)$, which is nonincreasing in N.